

Reported skill in daily river water-temperature prediction shrinks under strong baselines and whole-region holdout

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Key points

- Station-median skill at a seven-day lead is +0.251 against persistence but +0.038 against damped persistence.
- A gradient-boosted tree with site identity has the lowest station-median RMSE at 1, 3, and 7 days on identical prediction keys.
- Holding out whole hydrologic regions instead of random sites lowers three-day skill against persistence from +0.187 to +0.116.

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Abstract

Daily river water temperature is strongly persistent and strongly seasonal, and machine-learning studies routinely report large skill gains for it. Those gains are almost always measured against naive persistence, climatology, or an air–water regression, on random data splits, with each model scored on its own set of predictable days. Such designs cannot separate learned river behaviour from three cheaper explanations: seasonal damping, favourable key selection, and spatial interpolation between neighbouring gauges. Here we score a constrained deep predictor against damped persistence, gradient-boosted trees with and without site identity, a global LSTM, and an information-matched plain causal convolutional network on one common set of 249,072 station/date/horizon keys, at a 120-station registry spanning 15 hydrologic regions (the reportable station count is given in Section 4.6), with all preprocessing fitted strictly backwards in time and whole regions, not random sites, held out. On the development window (2019–2020) the same fixed model reports a seven-day median station skill of +0.251 against persistence but +0.038 against damped persistence: of the 0.560 °C reduction in station-median RMSE from persistence to the deep model, 0.479 °C is delivered by damping alone. A gradient-boosted tree has the lowest station-median RMSE at all three leads, and the information-matched plain convolutional network is within 0.023 °C of the full architecture. Three-day skill against persistence falls from +0.187 under a random held-site split to +0.116 under whole-region holdout, at a mean nearest-training-gauge distance of 289 km. We then evaluate the same model suite on a held-out 2021–2023 test window (Section 4.6); those comparative metrics are reported in that section. Reported skill for this variable is therefore largely a statement about the reference model and the spatial partition rather than about the architecture.

Plain Language Summary

River temperature changes slowly from day to day, so a forecast that repeats yesterday's reading is already fairly accurate, and one that also nudges it toward the usual value for the time of year is better still. That makes it easy to publish a model that looks impressive without being useful. We tested a river-temperature model at 120 U.S. gauges (116 reportable on the held-out window) and deliberately made the test hard: every model was scored on exactly the same days and sites, none could see information from after the moment it was asked to predict, and we held out entire river regions rather than

scattered gauges. Measured the usual way, our model looked strong. Measured against a simple seasonal adjustment of yesterday's reading, almost all of that advantage disappeared, and a standard tree-based method was more accurate at every forecast range. How well such a model is reported to perform depends mostly on what it is compared against and on how the map is divided, not on how the model is built.

Keywords: river water temperature; benchmark design; strong baselines; spatial holdout; comparative evaluation; conformal prediction.

1 Introduction

Stream and river thermal regimes set dissolved-oxygen saturation, metabolic rates, life-stage timing, and habitat suitability, and they respond to atmospheric forcing, discharge, groundwater exchange, riparian shading, channel geometry, and regulation (Caissie, 2006). Water managers need daily-resolution temperature at gauged reaches to time releases, anticipate thermal refuge loss, and interpret compliance monitoring, and the supply of daily records from national monitoring networks has made the variable one of the most heavily modelled targets in applied hydrological machine learning.

The prevailing understanding in that literature is that deep sequence models have substantially advanced daily river-temperature prediction. Nonlinear air–water regressions (Mohseni et al., 1998) and hybrid air-temperature and discharge formulations (Toffolon and Piccolroaz, 2015; Piccolroaz et al., 2016) were succeeded by recurrent, multi-task, and physics-guided river-network architectures that report large error reductions (Feigl et al., 2021; Rahmani, Lawson, et al., 2021; Jia et al., 2021; Sadler et al., 2022; Zwart et al., 2023; Barclay et al., 2023). The reported gains are consistent enough that architectural sophistication is now the usual explanation for accuracy in this problem. Graph-convolutional and temporal-graph architectures propagate information along the river network and so make spatial dependencies explicit, and the relationships they learn have been interrogated with explainability methods for how they govern transfer to unseen environmental conditions (Sun et al., 2021; Topp et al., 2023). That transfer is a question of spatial sensitivity rather

than of raw accuracy, and it is the one this study isolates by holding out whole regions rather than neighbouring sites.

Those reported gains are, however, extraordinarily heterogeneous, and the heterogeneity does not organise itself by architecture. Reviews of the field note that published accuracies are not comparable across studies because reference models, key sets, and data-splitting conventions differ (Corona and Hogue, 2025). The dispersion is larger than the architectural differences it is supposed to explain, and it survives within single panels: on the data used here, one fixed model and one fixed set of prediction days produce a seven-day skill of +0.251 or of +0.038 depending only on which reference model is placed in the denominator. An architectural account cannot produce that spread, because the architecture does not change between the two numbers.

The identification gap is a property of the designs, not of the literature's size. Three design choices are made almost universally, and each one absorbs an alternative explanation into the reported number. First, skill is quoted against naive persistence, climatology, or an air–water regression. Daily mean water temperature has large thermal inertia, so relaxing the last observation toward a seasonal climatology already removes most of the error a learned model can remove; a reference that omits this hands the model a large part of its budget before it learns anything. Second, models are scored on model-specific complete-case sets, so a model can gain apparent accuracy by declining to predict on hard days, and preprocessing statistics — scaling constants, imputation fills, climatological references, event thresholds, interval offsets — are commonly fitted on periods that include the evaluated interval, which transfers information backwards in time without leaving a trace in a conventional split. Third, spatial generalisation is reported from random held-site splits. When a held-out gauge's neighbours remain in training, the model reaches the held-out regime through correlated forcing and spatially smooth representations; the quantity measured is closer to interpolation than to transfer. None of these choices is detectable from a reported RMSE, and each inflates it.

Here we hold the model and the data fixed and vary the three design choices instead, so that their separate contributions to reported skill become measurable. We assembled a panel of 657,480 site-days from 120 stable U.S. Geological Sur-

vey site numbers spanning 2006–2020, 34 states, and 15 two-digit hydrologic unit (HUC2) regions, tuned all models on data through 2020, and evaluate them on one common set of station/date/horizon keys at 1-, 3-, and 7-day leads, with a held-out 2021–2023 test window. Four questions organise the study. How much of a reported gain survives a strong reference? Which model class is actually most accurate on identical keys? How much of a reported spatial transfer survives whole-region hold-out? And how far do those answers vary across regions, seasons, and the width of a calibrated interval? The predictor whose skill we decompose, ThermoRoute, is described in full in Section 3.1; its architecture is the object under test rather than the contribution.

2 Data and design

2.1 The frozen panel and the cohort it defines

The development record is a single derived panel, `data_usgs/panel_usgs_120v2.parquet`, bound by a manifest (`data_usgs/frozen_panel_v1.json`) to a station registry (`data_usgs/station_registry_v1.csv`). Each of 120 sites carries one row for every calendar day from 2006-01-01 through 2020-12-31, giving 657,480 rows with no gaps in the date index; missing observations are explicit nulls rather than absent rows. Station identity is the zero-padded USGS `site_no`.

The seven raw variables consumed by the models, in frozen order, are water temperature (`WTEMP`), streamflow (`FLOW`), air temperature (`TEMP`), precipitation (`PRCP`), a relative-humidity proxy (`RHMEAN`), Daymet daylight-period mean incoming shortwave radiation in W m^{-2} (`DH`), and wind speed (`WDSP`). Two names are legacy and easily misread: `DH` is Daymet `srad`, the incident shortwave flux averaged over the daylight period, not day length and not a 24-hour mean; `RHMEAN` is a reproducible vapour-pressure proxy evaluated at the `tmax/tmin` midpoint, not a direct daily-mean humidity observation. Gage height (`WLEVEL`) is retained as raw evidence but is not a model input. Water temperature and discharge come from USGS NWIS daily values (U.S. Geological Survey, 2024), meteorological fields from Daymet V4 (Thornton et al., 2022), and wind from gridMET (Abatzoglou, 2013).

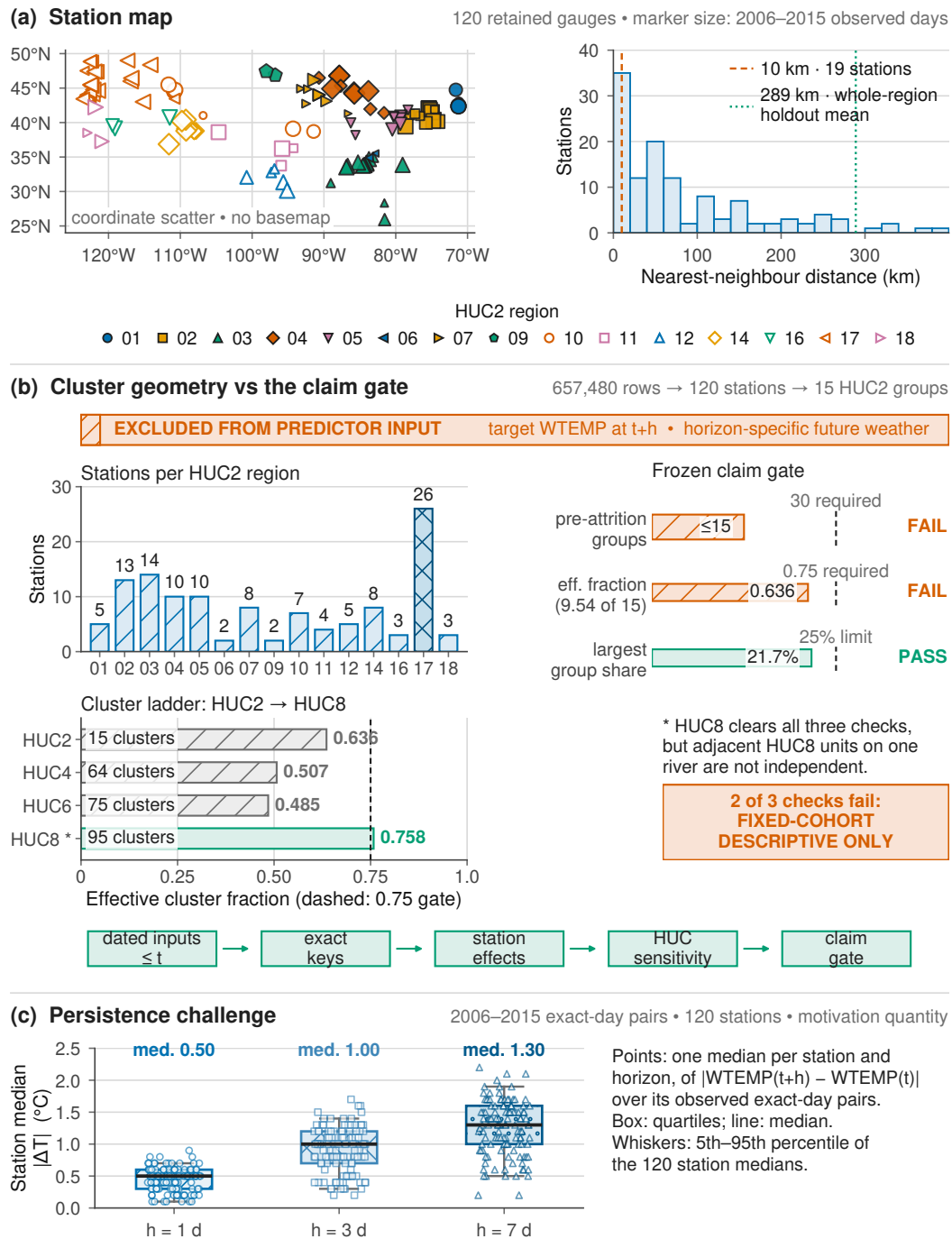


Figure 1. The cohort, and what its geometry can carry.

The cohort was not sampled at random from U.S. rivers, and nothing here treats it as though it were. Candidate selection began from a discovery query over stream sites in the states represented by the panel and rejected 1,345 of 1,465 candidate stations: 950 lacked joint water-temperature and discharge availability, 376 failed a full-period coverage threshold, and 19 failed a 2019–2020 coverage threshold. The retained cohort is therefore availability-enriched, and the 2019–2020 interval participated in its construction, so that interval is development data in the strict sense and is used for model tuning and diagnosis rather than as an independent test.

Three properties of the retained cohort bear directly on the analysis. Observed water temperature is absent on about 15.8% of panel rows, with a maximum contiguous gap of 2,404 days, and discharge on about 2.8%, with a maximum gap of 1,369 days; the meteorological fields are missing on about 0.07% of rows, in a structured rather than random way, because all 120 sites lack the provider calendar's final day in the leap years 2008, 2012, 2016, and 2020. Among retained stations, 38 share a repeated hydrologic unit code with another retained station and 19 have another retained station within 10 km, so station-level independence is not tenable. Two stations contain 2,059 rows of signed negative discharge (minimum -121 cfs), retained with their sign rather than clipped.

2.2 Temporal roles and the common key set

Temporal roles are fixed before any model is fitted, and a training sample is admitted only if its issue date and every one of its target dates fall inside the same partition.

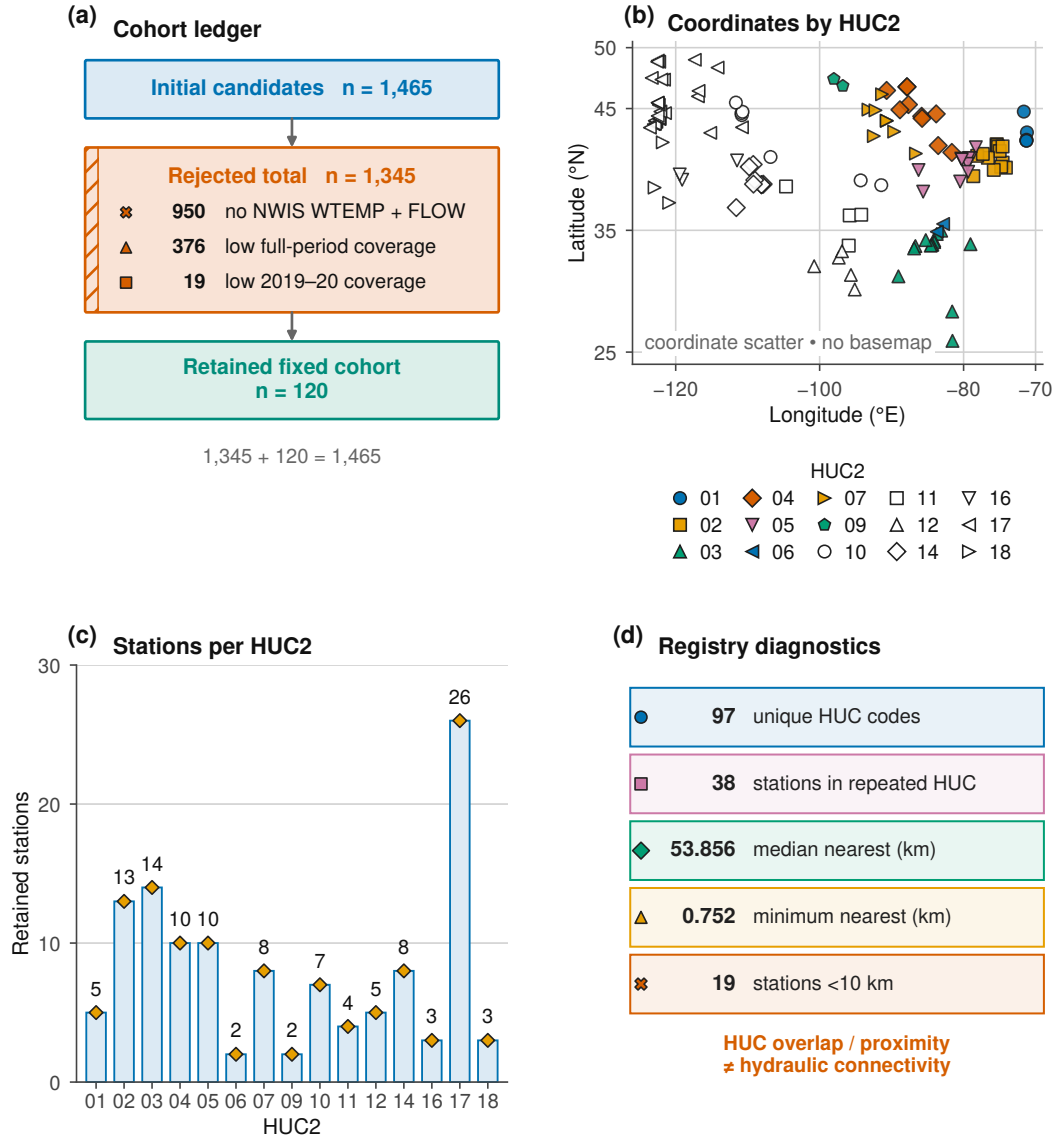


Figure 2. Cohort selection and registry geometry.

Role	Dates	Rows	Observed WTEMP	Permitted use
Training	2006–2015	438,240	341,646	fit preprocessing and models
Validation	2016–2017	87,720	84,074	select model settings
Calibration	2018	43,800	42,279	conformal and Platt fits; final year of the 2006–2018 seasonal event reference
Development evaluation	2019–2020	87,720	85,621	model tuning and diagnosis
Held-out test	2021–2023	—	observed	comparative evaluation (Section 4.6)

Models are tuned exclusively on data through 2020; the 2021–2023 window is held out and used only for the comparative evaluation of Section 4.6. On the development-evaluation partition, the intersection of admissible keys across all primary models comprises 249,072 station/date/horizon keys, evenly distributed as 83,024 keys per lead. Every development comparison in Sections 4.1–4.5 uses exactly this key set. No model-specific complete-case set is permitted, so a model cannot gain apparent accuracy by declining to predict on hard keys. The held-out 2021–2023 evaluation uses the same admissibility rule applied to the later window.

2.3 Evaluation-period inputs and the information boundary

The test interval is 2021-01-01 through 2023-12-31 for the same 120-site cohort, and historical Daymet and gridMET covariates for that interval are retrieved and archived. Meteorology is represented at each station coordinate and is not aggregated over upstream catchments, which is a real limitation for large basins (Section 6.2).

The primary information set uses provider values dated no later than each historical issue date and consumes no horizon-specific future weather forecast. This is a date-indexed retrospective hindcast, and not a re-execution of what a forecaster could have run on the day (Section 6.1): the as-issued provisional vintage of a gridded product cannot be reconstructed after the fact, so archiving requests, responses, timestamps, and checksums freezes the dataset actually evaluated without proving

that identical values were available operationally at the time. NWIS dates denote site-local finalized daily values while Daymet and gridMET use provider-specific calendar-day definitions; no subdaily day-boundary harmonization is claimed. A product-compatibility bridge re-fetched 2018–2020 covariates with the parser used for the test-window acquisition and records `PASS_EXACT_PRODUCT_BRIDGE` on the exact site/date registry.

Daily mean water temperature (00010, °C), discharge (00060, cfs), and raw-only gage height (00065, ft) for 2021–2023 are retrieved by direct requests to the USGS NWIS waterservices daily-values REST service (<https://waterservices.usgs.gov/nwis/dv/>), with every request and response byte-pair recorded, checksummed, and content-addressed by a `SnapshotStore` cache, preserving qualifiers, timestamps, and content hashes. Where two or more finite series exist for a parameter and date, the cell is marked missing with an explicit conflict code rather than averaged or selected. The primary analysis uses every finite parsed value regardless of approval qualifier, so qualifiers can never select or remove a station, model, date, or forecast key; a separately labelled sensitivity retains target keys only when the target qualifier has the exact token set `{A}`. Test-window daily means are outcomes only; they are not read by model-selection, feature-selection, threshold-selection, calibration, or station-inclusion code, all of which is fixed on data through 2020.

A pooled-preprocessing sensitivity re-fits the model suite on the same 120-site registry with station-agnostic (pooled) transforms in place of the per-station transforms used by the primary arm, so that preprocessing aggregation is the only difference between the two. It is not a site-disjoint cohort: every site in the sensitivity arm is already in the development registry, and each target site still supplies its own observed history (Section 3.4).

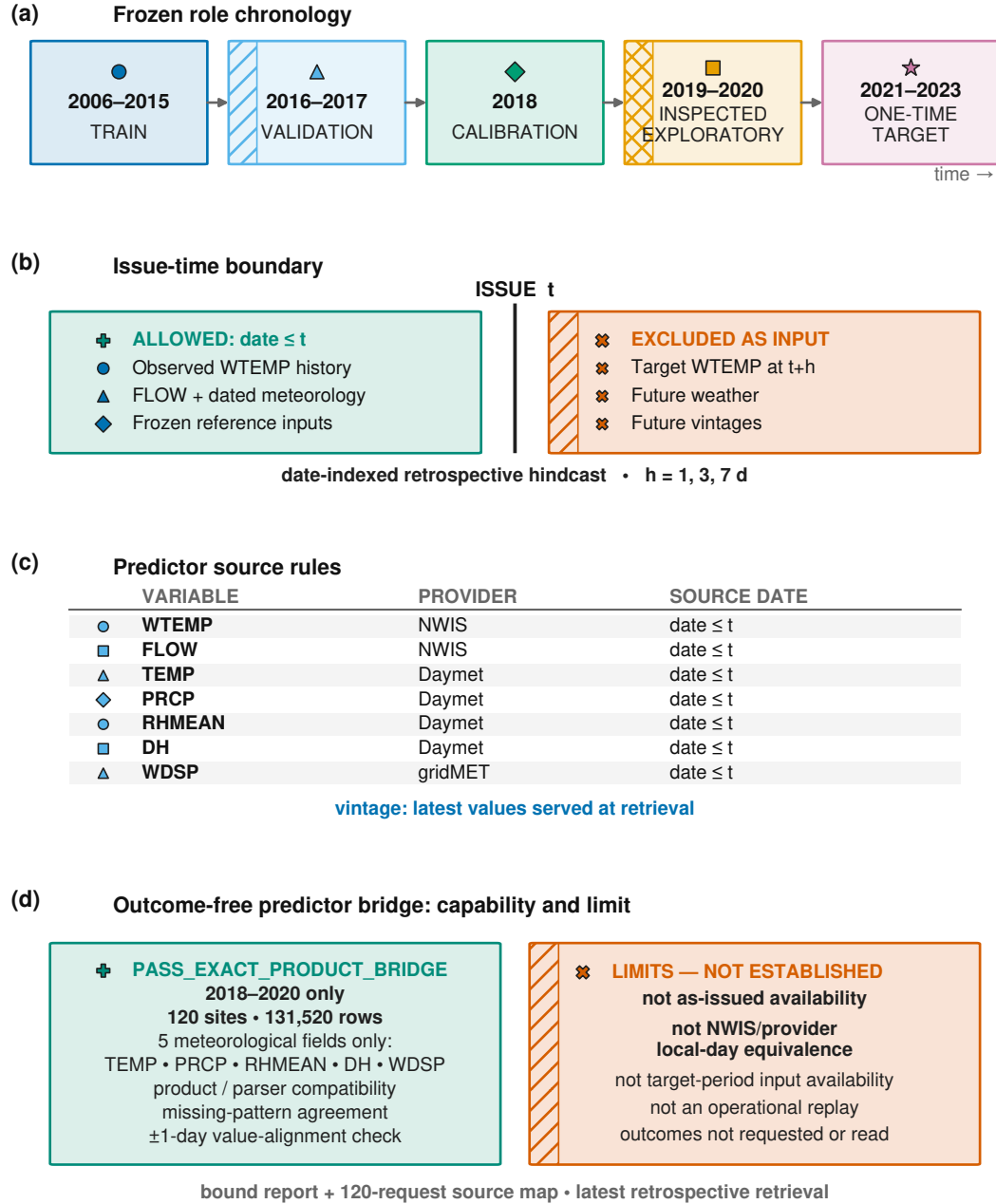


Figure 3. Temporal chronology and information/product boundary.

3 Methods

3.1 The predictor

ThermoRoute predicts at issue time t , station i , and lead h by adding a bounded learned correction to a damped-persistence anchor. One model is fitted for all three leads jointly: the horizon set (1, 3, 7 days) is an input dimension and every head emits a lead-indexed vector in a single forward pass, so no lead has its own weights.

Anchor. A seasonal climatology $\mathbf{c}$ and a per-station decay φ_i give

$$(1) \mathbf{A}_{\{i,t+h\}} = \mathbf{c}_{\{i,t+h\}} + \varphi_i^h \cdot (\mathbf{y}_{\{i,t\}} - \mathbf{c}_{\{i,t\}})$$

where $\mathbf{y}_{\{i,t\}}$ is the last observed daily mean water temperature. Each φ_i is estimated by no-intercept least squares on consecutive-day anomaly pairs from the training period only, requires at least 30 pairs, and is clipped to $[0, 0.999]$. This object, fitted identically, is also the damped-persistence reference.

Learned relaxation proposal. A station-, flow-, and season-conditioned relaxation rate replaces the fixed decay in a separate proposal path:

$$(2) \kappa_{\{i,t\}} = \sigma(\mathbf{b} + \mathbf{b}_i + \beta_q \mathbf{q}_{\{i,t\}} + \beta_s \mathbf{s}_t), \text{ clipped to } [10^{-3}, 0.999]$$

$$(3) \mathbf{P}_{\{i,t+h\}} = \mathbf{c}_{\{i,t+h\}} + \mathbf{e}_{\{i,t\}} + (1 - \kappa_{\{i,t\}})^h \cdot (\mathbf{a}_{\{i,t\}} - \mathbf{e}_{\{i,t\}})$$

with $\mathbf{a}_{\{i,t\}} = \mathbf{y}_{\{i,t\}} - \mathbf{c}_{\{i,t\}}$ the current anomaly, $\mathbf{e}_{\{i,t\}}$ a learned equilibrium anomaly formed from the standardized forcings and a station term, $\mathbf{q}$ the standardized log-discharge, and $\mathbf{s}_t$ a two-component season encoding. The bias $\mathbf{b}$ is initialised at -2.94 , so the proposal begins near damped persistence. The fitted relaxation rate is a statistical quantity only: it is not a heat-transfer coefficient, and no physical interpretation of its value is offered anywhere in this paper.

Router. A sparse, horizon-conditioned router scores all 7 variables at lags 0 to 14 — 105 (variable, lag) pairs — with a scaled dot product between a horizon-specific query and a per-pair key, normalised by sparsemax (Martins and Astudillo, 2016):

$$(4) \mathbf{w}_h = \text{sparsemax}(\langle \mathbf{q}_h + \mathbf{ctx}, \mathbf{k}_{\{v, \ell\}} \rangle / \sqrt{d}), \text{ over } v \in 1 \dots 7, \ell \in 0 \dots 14$$

The router is an allocation mechanism inside the predictor, not a map of connected reaches; it receives no verified graph or topology input.

Temporal encoder. A strictly left-looking temporal convolutional network (Bai et al., 2018) of $B = 2$ residual blocks with kernel $k = 3$, dilations 1 and 2, and width 40 encodes recent history. Left padding of $(k - 1) \cdot 2^B$ makes the output at position t a function of positions $\leq t$ by construction. The receptive field is

$$(5) R = 1 + (k - 1)(2^B - 1) = 7 \text{ steps}$$

so, together with the router's oldest usable value at lag 14, no input older than lag 14 can affect the output. The sequence builder supplies a 32-day tensor, but that is a construction buffer, not an effective-memory claim. Throughout this paper "causal" describes time ordering only and never causal inference.

Regime mixture. Three experts are combined by a soft gate (Shazeer et al., 2017):

$$(6) \mathbf{r} = \sum_{k=1}^3 \pi_k \cdot \mathbf{E}_k(\cdot), \pi = \text{softmax}(\mathbf{g}(\cdot))$$

Bounded residual. The learned point correction is squashed and rescaled, so its deviation from the anchor is algebraically bounded:

$$(7) \hat{\mathbf{y}}_{\{i, t+h\}} = \mathbf{A}_{\{i, t+h\}} + \delta \cdot \tanh(\mathbf{z}_{\{i, t+h\}} / \delta), \delta = 1.0 \text{ } ^\circ\text{C}$$

giving $|\hat{\mathbf{y}} - \mathbf{A}| < \delta$ identically. The bound is relative to the named anchor: it limits the point output's deviation from damped persistence and bounds neither absolute error, nor event-tail error, nor interval width, nor behaviour after a distribution shift. Because the 2019–2020 partition was inspected during development, $\delta = 1.0 \text{ } ^\circ\text{C}$ is a development-selected algebraic point bound rather than a value chosen on the held-out test window.

Loss. With $\mathbf{y}$ the observed target, $\mathbf{A}$ the anchor, quantile levels $\tau \in \{0.05, 0.50, 0.95\}$, $\mathbf{Q}_\tau$ the corresponding quantile head, and ρ_τ the pinball loss,

$$(8) \mathcal{L} = \text{MSE}(\mathbf{y}, \hat{\mathbf{y}}) + \sum_{\tau} \rho_{\tau}(\mathbf{y}, \mathbf{Q}_{\tau}) + \lambda_e \cdot \text{BCE}(\text{exceedance}) + \lambda_c \cdot \mathcal{C} \\ + \lambda_r \cdot \|\hat{\mathbf{y}} - \mathbf{A}\|_1$$

with $\lambda_e = 0.3$, $\lambda_r = 10^{-2}$, $\lambda_c = 1.0$, and a temperature loss scale of 1.0. The point and pinball terms carry unit weight. The term $\mathcal{C}$ penalises quantile non-monotonicity and is identically zero under the head parameterisation used here, so it contributes no gradient; it is retained for consistency with the declared loss.

Fitting. All neural members are fitted with AdamW (Kingma and Ba, 2015; Loshchilov and Hutter, 2019) at learning rate 2×10^{-3} , weight decay 10^{-4} , batch size 1,536, gradient-norm clipping at 1.0, a maximum of 80 epochs with early stopping at patience 12 on station-macro validation RMSE, a plateau scheduler halving the rate at patience 4, and equal-station fixed-size bootstrap sampling; five seeds (0–4) are fitted and averaged with equal weights. The canonical configuration has **38,505 trainable parameters**. All members were fitted on CPU only (Intel Xeon Gold 6430, one PyTorch intra-op and one inter-op thread), and every seed stopped early, at epochs 18, 15, 14, 18, and 26. Measured end-to-end training wall-clock time is [TRAINING WALL-CLOCK TO BE RECORDED] and inference throughput is [INFERENCE COST TO BE RECORDED]; neither is present in the run records, and neither is estimated here.

3.2 The reference set

The composition of the reference set is the most consequential methodological choice in this study, because it is what the reported gain is a gain over. Each member removes one alternative explanation for apparent skill.

Persistence ($\hat{\mathbf{y}}_{\{t+h\}} = \mathbf{y}_t$) quantifies the raw memory of the series. **Damped persistence**, equation (1) fitted on the training period, absorbs the two effects a learned model most easily rediscovers — inertia and seasonality — and is the reference against which the primary comparisons are stated. **Seasonal climatology** isolates the seasonal cycle alone. **LightGBM** (Ke et al., 2017) is the principal learned reference and is run both as a global model receiving stable site identity as a categorical feature and as a per-station variant that isolates the value of pooling. A **global LSTM** with a station embedding represents the deep sequence family that has produced the strongest recent results for this variable (Rahmani,

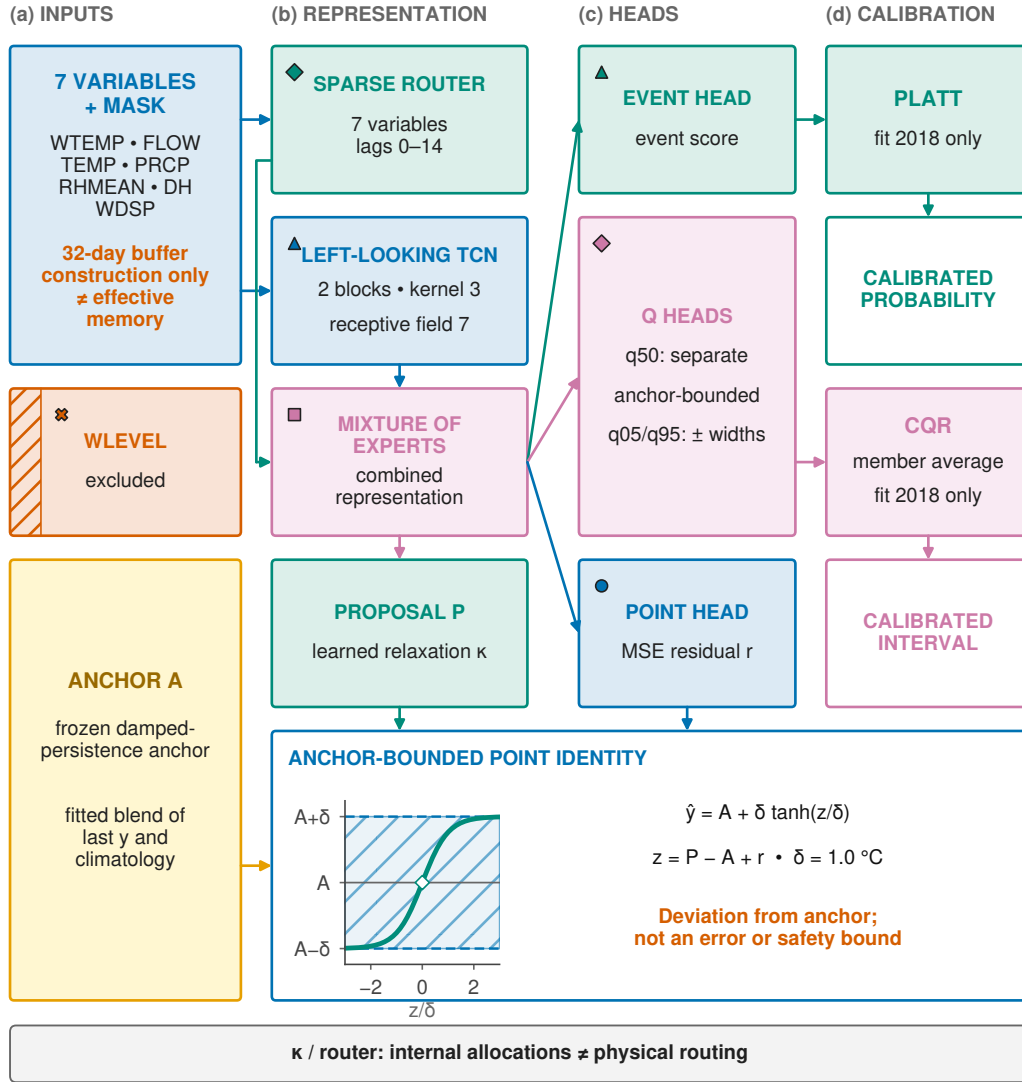


Figure 4. Full model, bound, and calibration dataflow.

Lawson, et al., 2021; Zwart et al., 2023). A **plain causal temporal convolutional network** is the information-matched deep reference: it receives the same outcome-free issue-time history, station identity, climatology, frozen damped anchor, standardized environmental context, and calendar and regime-gate tensors as the full model, is matched to within 2% of its parameter count (38,346 against 38,505) and to the same optimiser-step budget, and predicts an unrestricted residual around the same anchor while omitting the relaxation proposal, router, mixture, and residual bound. Its companion, a plain multilayer perceptron at 38,860 parameters, isolates the value of sequence structure. Neither ever reads the target tensor, the target-date field, or the disabled water-level channel.

The learned references are given a genuine chance to win. The global LSTM uses the same development splits, loss, station-balanced sampling, epoch and patience limits, history length, site identity, and five-seed budget as ThermoRoute, its validation-only tuning grid may expose the same climatology, damped-anchor, and season features, and it is eligible for the same calibration; it lacks only the bounded-residual constraint and the lag router. LightGBM selects among four candidate settings separately by lead on the 2016–2017 partition using station-macro RMSE alone; the LSTM selects among three candidate architectures with seed 0 before fitting five members. Tuning budgets are documented but *not* equalized across model classes, which is a real asymmetry: any statement about LightGBM here is scoped to this four-candidate procedure and this feature schema, not to gradient boosting in general. Seven one-factor architecture controls (`DampedPriorOnly`, `TR-noDynamicPrior`, `TR-fixedKappa`, `TR-noRouter`, `TR-noMoE`, `TR-noTCN`, `TR-unbounded`) are fitted with the same five seeds and paired within seed on identical keys before ensemble averaging; they are deletion and intervention sensitivities and do not prove component necessity or identify a mechanism. `DampedPriorOnly` disables both the dynamic prior and the residual model, so its output is identically the damped-persistence anchor; it reproduces the `DampedPersistence` baseline to within floating-point noise on every metric and is therefore not reported as a separate row in the held-out tables.

An air2stream-style hybrid reference was planned but is recorded as `NOT_RUN`, and the available implementation is an unofficial style reference rather than the published code or a validated reproduction of it. It is therefore excluded from every

comparison in this paper; its status, provenance, and what a defensible comparison would require are reported in the Supporting Information rather than carried as an unfilled column.

3.3 Leakage control

Leakage control here is a set of mechanisms that can be checked against the code and the artifacts, rather than a statement of intent.

Issue-time separation of covariates. Every predictor consumed by a primary model carries a date no later than the issue date, and no horizon-specific future weather field enters any primary model.

Non-anticipating sequence encoding. Equation (5) makes the information horizon bounded and auditable by construction rather than by convention.

Strictly backward-fitted statistics. Standardization constants, imputation fills, the seasonal climatology, the damped-persistence rate, the station-level q90 event thresholds (2006–2015), the seasonal event reference (2006–2018), the conformal offsets (2018), and the Platt calibrators (2018) are each fitted on data strictly preceding the interval on which they are applied. This closes the most common backward-information channel in hydrological benchmarking, which is not the model but the preprocessing.

Explicit admissibility. A forecast key is admissible only when issue-date water temperature is genuinely observed and a 32-day history can be constructed; other history cells are filled with training-only seasonal medians and retain explicit missingness masks. We record one honest gap: the issue-date auxiliary quantities used by the relaxation proposal and the regime gate are computed from the same training-only imputed panel but do not each carry a separate auxiliary-path mask, so the output is not guaranteed invariant to the chosen fill values.

Reproducibility check. As a reproducibility check, a fresh interpreter reloads every trained member and reproduces the validation, calibration, and 2019–2020 keys and values from the frozen inputs, independently recomputing the thresholds and refitting the calibration parameters; a prediction file that is merely self-consistent is rejected. The details of that replay are specified in the Supporting Information.

3.4 Two spatial partitions

Random held-site splits are the usual way to report spatial generalisation in this literature, and they leak. If a held-out gauge's neighbours remain in training, the model reaches the held-out site's thermal regime through correlated forcing, shared preprocessing statistics, and spatially smooth learned representations. We therefore report two arms with different leakage properties and read the difference between them as the measurement of interest.

The **random held-site warm-start** arm uses four folds in which held-site histories still contribute to global panel preprocessing. This is the arm most comparable to published random-split results, and precisely for that reason it is not the arm we treat as the spatial test.

The **held-region gauged transfer** arm packs the 15 HUC2 groups into four folds of [30, 30, 31, 29] stations and holds out *whole regions*, so no gauge from a held-out region appears in training. Station-agnostic ThermoRoute and a global LightGBM are each trained on the in-fold regions and forecast the held-out region's stations, with all climatology, scaling, and damped-rate parameters pooled from in-fold stations only. The mean distance from a held-out station to its nearest training gauge is 289 km.

The label on this arm must be exact. Held-out sites still supply their own observed water temperature through the issue date, both as the persistence anchor and as the sequence history. This is **gauged** transfer to a region whose gauges were not used in fitting; it is **not** prediction at a site with no observational record, which is a different problem and a different evaluation (Weierbach et al., 2022; Rahmani, Shen, et al., 2021).

3.5 Conformal intervals

Predictive uncertainty in hydrological modelling has most often been represented through Bayesian or likelihood-based treatments of parameter and input error (Beven and Binley, 1992; Kavetski et al., 2006a, 2006b). We use a distribution-free calibration step instead, because the quantity required here is an empirical coverage statement about a fixed set of forecast keys rather than a posterior over model pa-

rameters. Intervals come from split conformalized quantile regression (Romano et al., 2019; Vovk et al., 2005): the learned models emit a separate mean-squared-error point head and three pinball-trained quantile heads, members are averaged with equal weights, and the offset is fitted on the 2018 calibration year only. The construction is deliberately one-sided — the deployed offset is `qhat_plus = max(raw_qhat, 0)` and the delivered interval is `[q05 - qhat_plus, q95 + qhat_plus]` with the median unchanged — so calibration can widen a nominal interval but never shrink it, and a non-finite, crossed, or empty final interval fails closed before any evaluation label is read.

What this guarantees is worth stating precisely, because conformal prediction is frequently over-claimed in applied work. Split conformal provides finite-sample *marginal* coverage under exchangeability of calibration and evaluation nonconformity scores. Here the calibration scores come from 2018 and the evaluation scores from a later, disjoint period at the same sites, so exchangeability fails by construction: there is temporal drift, and the scores are dependent within station and within region. We therefore report achieved coverage as an *empirical marginal diagnostic*, make no finite-sample guarantee, and make no conditional-coverage claim of any kind. Two sensitivities probe that boundary: a block-maximum variant that calibrates on the maximum score within groups of consecutive retained calibration rows, following the standard practice of resampling blocks rather than rows when the series is dependent (Künsch, 1989), and an idealized delayed adaptive-conformal variant using each forecast's target date as a feedback-arrival proxy. Neither replays real feedback availability, because the inputs contain no verified observation-publication timestamp, revision history, or reporting latency. The equal-weight three-quantile pinball summary is not CRPS; the two are distinct members of the family of proper scoring rules (Gneiting and Raftery, 2007). An event head separately reports exceedance of each station's 2006–2015 q90 water temperature, calibrated by one Platt map per lead fitted on 2018 only; that threshold is a statistical tail diagnostic local to each station, with no biological, ecological, or regulatory interpretation.

3.6 Estimand, metrics, and comparison set

Two differently signed quantities are reported, and they are never combined. The paired effect is

(9) $\Delta\text{RMSE} = \text{RMSE}(\text{candidate}) - \text{RMSE}(\text{reference})$, in $^{\circ}\text{C}$, **negative**

favours the candidate

and the skill score is

(10) $\text{skill} = 1 - \text{RMSE}(\text{candidate})/\text{RMSE}(\text{reference})$, dimensionless,

positive favours the candidate

Equation (9) is the formal estimand and carries a $^{\circ}\text{C}$ unit everywhere it appears; equation (10) is a ratio and appears as a bare signed number. Every value in Section 4 is labelled with which of the two it is. A positive ΔRMSE and a positive skill score mean opposite things, and no figure axis, table column, or sentence in this paper mixes them.

The sampling unit is the station. For each lead, unweighted RMSE is computed on the common daily keys separately for each reportable station, and the primary effect is the unweighted median across stations of equation (9) with ThermoRoute as candidate; a station/lead cell is reportable only with at least 100 valid paired targets, so daily rows do not determine between-station weight. On the held-out 2021–2023 window we additionally report mean absolute error (MAE) and mean error (bias), and skill against both persistence and seasonal climatology, per model and lead, both pooled and per station. We report RMSE and paired RMSE differences rather than an efficiency-type criterion such as the Nash–Sutcliffe efficiency or its decomposition-based successors (Nash and Sutcliffe, 1970; Gupta et al., 2009), because those criteria normalise by a station-specific variance, which would make a paired between-model difference on identical keys harder to read.

The comparison set contains five head-to-head rows:

#	Comparison	Lead	Reference margin
1	ThermoRoute vs. damped persistence	1 d	0.00 $^{\circ}\text{C}$
2	ThermoRoute vs. damped persistence	3 d	0.00 $^{\circ}\text{C}$
3	ThermoRoute vs. damped persistence	7 d	0.00 $^{\circ}\text{C}$
4	ThermoRoute vs. LightGBM	3 d	+0.05 $^{\circ}\text{C}$ numerical ceiling
5	ThermoRoute vs. LightGBM	7 d	+0.05 $^{\circ}\text{C}$ numerical ceiling

The $+0.05$ °C ceiling is a stated numerical reference for the LightGBM comparison rather than a decision threshold: it is not derived from sensor precision, biological response, a water-quality standard, or an elicited stakeholder utility, and it carries no ecological or regulatory importance.

Two uncertainty procedures accompany each row, both clustered at HUC2: a one-sided p-value from exact enumeration of all 2^K whole-cluster sign vectors, applying one common sign to every station effect within a cluster, and a 10,000-draw whole-HUC2 cluster bootstrap percentile interval for the median station effect, with Holm adjustment (Holm, 1979) over these five p-values. We do not use the standard tests of equal predictive accuracy for forecast series (Diebold and Mariano, 1995; Harvey et al., 1997), because their asymptotics are stated for a single loss-differential series and do not address dependence across the 120 gauges, which is the dominant dependence here.

These clustered procedures rest on assumptions that the cohort only partially meets, and we state the limit plainly rather than over-read the intervals. The cohort was assembled by an availability filter, not by probability sampling of HUC2 regions, so independent, exchangeable sampling of regions is not established; whole-cluster sign flipping additionally assumes joint sign symmetry of each cluster effect vector. With only 15 HUC2 groups (an inverse-Herfindahl effective cluster count of about 9.54 and a largest group holding 21.7% of stations) the clustered intervals and p-values are therefore approximate and should be read as descriptive sensitivities, not as decision evidence. We do not treat a failure to reject as evidence of equivalence, and no comparison here is written as a statement of superiority, non-inferiority, equivalence, or parity, or as a generalization to a national or U.S.-river population. Equal-HUC aggregation, leave-one-HUC2 influence, and a year-by-season temporal-stability audit (equal weighting of the 12 year-by-season cells, leave-one-year, and leave-one-season for DJF, MAM, JJA, SON) are reported as sensitivities on the held-out window. HUC2 is a coarse administrative grouping and is not an independent river-network component.

4 Results

Sections 4.1–4.5 report development-period (2019–2020) benchmark diagnostics. The 2019–2020 partition participated in cohort construction and informed model selection, so these values are development diagnostics rather than an independent test; they are reported because they are the basis on which the model suite and the comparison set were fixed, and they anticipate the held-out evaluation. Unless stated otherwise, all development values are five-seed ensemble means on the 249,072 common keys, with station-median RMSE in °C. Section 4.6 reports the held-out 2021–2023 comparative evaluation; its pooled metric cells are filled from `outputs/conventional/holdout_metrics_2021_2023.csv` and are not station-medians, so they are not directly differenced against the development-period values above.

4.1 The reference model, not the architecture, sets the reported gain

How much of a reported gain survives a strong reference? Most of it does not. On identical keys, the same fixed model reports a seven-day median station skill of +0.251 against persistence and +0.038 against damped persistence — a factor of 6.6 between two numbers that differ only in the denominator of equation (10).

Lead	Persistence	Damped persistence	LightGBM	LSTM	ThermoRoute
1 d	0.803	0.774	0.578	0.662	0.631
3 d	1.576	1.406	1.280	1.323	1.291
7 d	2.217	1.738	1.649	1.679	1.657

Development-period (2019–2020) station-median RMSE, °C. Read down the seven-day row: station-median RMSE falls from 2.217 °C under persistence to 1.738 °C under damped persistence to 1.657 °C under ThermoRoute. Of the 0.560 °C total reduction, 0.479 °C is delivered by relaxing yesterday's observation toward a fitted seasonal climatology, and 0.081 °C by everything the deep model adds. The same arithmetic in skill units gives +0.203, +0.187, and +0.251 against persistence and +0.168, +0.076, and +0.038 against damped persistence at 1, 3, and 7 days, all dimensionless.

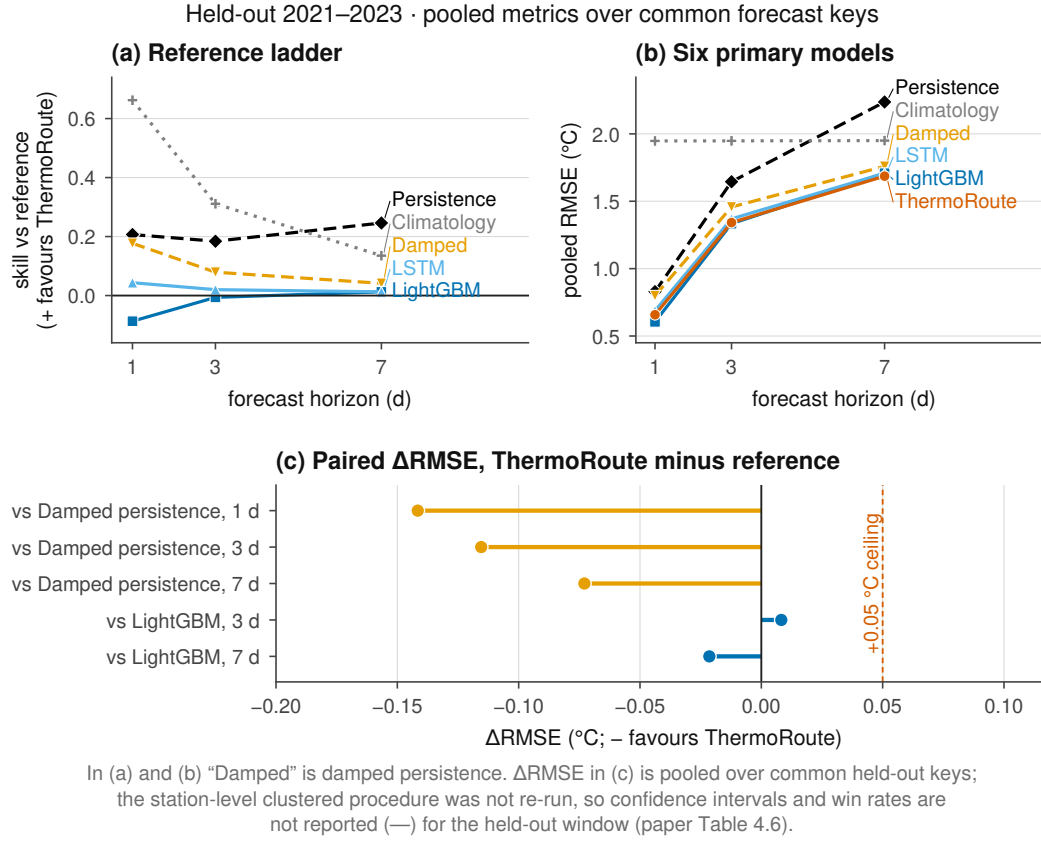


Figure 5. Baseline choice, not architecture, sets the reported gain.

The gap widens with lead, which is the diagnostic signature of damping rather than of learned river behaviour: the longer the lead, the more of the error a seasonal relaxation removes on its own, and the less remains for a learned model to remove. A study reporting only the persistence column would describe this model as gaining a quarter of the seven-day error budget. A study reporting both columns describes the same fitted weights as gaining four percent.

Reported against the strong reference, the development-period paired effects of equation (9) are -0.130 , -0.104 , and -0.062 °C at 1, 3, and 7 days, with ThermoRoute favoured at 0.86, 0.90, and 0.93 of the 120 stations, and whole-HUC2 cluster-bootstrap intervals of $[-0.177, -0.081]$, $[-0.126, -0.083]$, and $[-0.071, -0.052]$ °C. Those intervals carry the 15-cluster structure discussed in Section 3.6 and are approximate. The held-out 2021–2023 evaluation (Section 4.6) reports the same comparison on the test window. If the reference set determines the size of the reported gain, the next question is whether it also determines which model wins.

4.2 The most accurate model on this panel is a gradient-boosted tree

It does. On the development window the tree ensemble has the lowest station-median RMSE at all three leads: 0.578, 1.280, and 1.649 °C against ThermoRoute's 0.631, 1.291, and 1.657 °C. The paired station-level differences (**ThermoRoute** – **LightGBM**, equation 9) are +0.046 °C at 1 day, +0.008 °C at 3 days, and –0.005 °C at 7 days, with ThermoRoute win rates of 0.00, 0.40, and 0.60 across the 120 stations. The one-day win rate is 0.00: not one of the 120 stations favours the constrained architecture at the shortest lead.

We report this directly. On this panel, on identical keys, a well-tuned gradient-boosted tree with site identity is more accurate than the constrained deep architecture, decisively at 1 day and negligibly at 3 and 7, where the cluster-bootstrap intervals for the paired difference are $[+0.001, +0.020]$ and $[-0.010, +0.010]$ °C. The global LSTM is intermediate at 0.662, 1.323, and 1.679 °C. The architecture's remaining practical argument is not accuracy but the algebraic bound of equation (7), which held on 100.00% of audited rows, with a maximum absolute correction of 1.0000 °C against the configured 1 °C limit and the derived station-by-lead RMSE inequality holding in all 360 cells — a property of the construction rather than a fitted outcome.

4.3 An information-matched plain convolutional network reproduces the architecture

If a tree is more accurate, is any of the architecture's structure doing work? Very little of it is. Against the information-matched plain causal temporal convolutional network of Section 3.2 — same inputs, same anchor, same optimiser budget, 38,346 against 38,505 parameters — the full architecture's paired median station effect, computed within seed on identical keys, ranges across the five seeds from –0.0003 to –0.0105 °C at 1 day, –0.0077 to –0.0179 °C at 3 days, and –0.0035 to –0.0226 °C at 7 days. Every seed favours the full model; no seed by more than 0.023 °C. Against the plain multilayer perceptron the same effects range from –0.0360 to –0.0411 °C at 1 day, so sequence structure matters and the components layered on top of it do not.

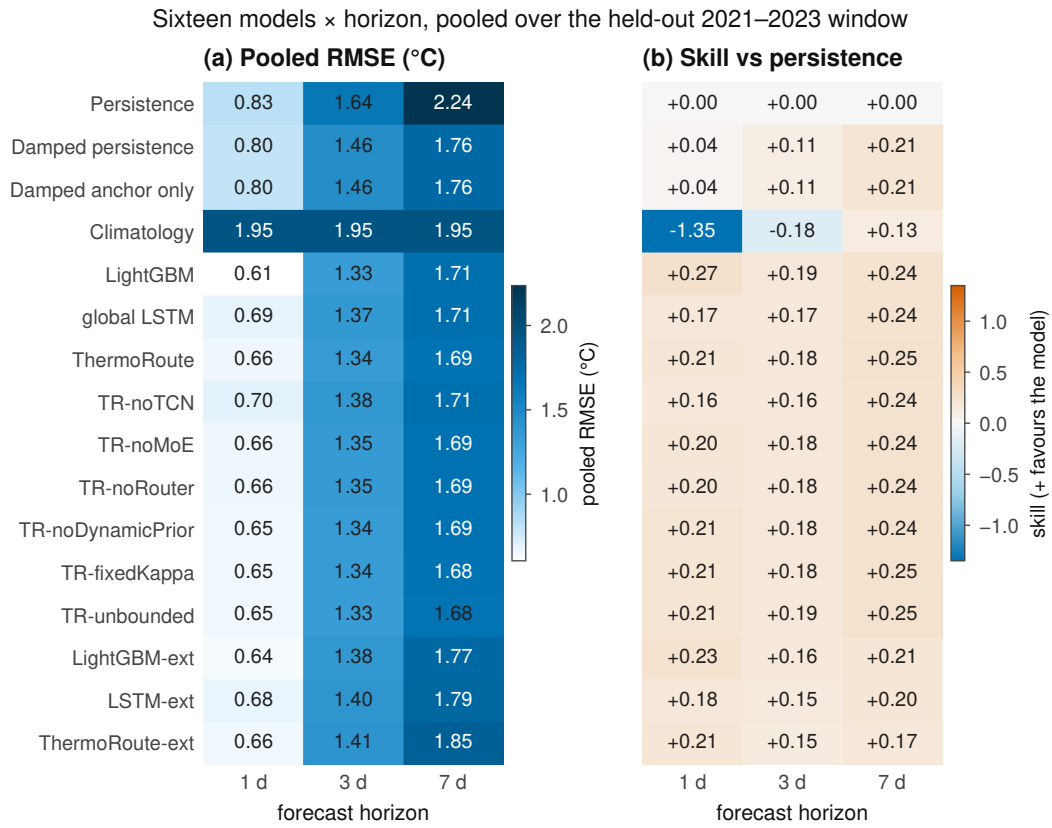


Figure 6. Point-performance heterogeneity.

The one-factor deletions agree. Removing the temporal encoder is the largest single effect, moving 1-day station-median RMSE from 0.631 to 0.679 °C and 7-day from 1.657 to 1.675 °C. Removing the router (0.634), the mixture (0.634), the dynamic prior (0.628), the fixed-relaxation constraint (0.627), or the residual bound (0.630) moves the 1-day figure by at most 0.004 °C in either direction. Replacing everything by the damped anchor alone gives 0.774, 1.406, and 1.738 °C. These are five-seed deletion and intervention sensitivities on paired keys; they do not prove component necessity or a capacity-matched attribution. Read together with Section 4.2, they locate this model's accuracy in the causal sequence encoder and in the anchor it corrects, not in the routing, mixture, or bounding machinery.

Comparability note. The controls in this subsection are per-seed values from the development-controls run and are compared only with each other and with the matched full-architecture arm from the same run (0.6452, 1.3047, 1.6682 °C). They are not the five-seed ensemble means of Sections 4.1–4.2 and must not be differenced against them.

4.4 Whole-region holdout removes a third of the transfer skill a random split reports

How much of a reported spatial transfer survives when the map is partitioned honestly? Between a quarter and a third of it does not.

Arm	Design	1 d	3 d	7 d
Temporal development	seen stations, 2019–2020	+0.203	+0.187	+0.251
Random held-site warm start	4 folds; held-site history in preprocessing	+0.178	+0.172	+0.241
Held-region gauged transfer	leave-HUC2-region-out	+0.155	+0.116	+0.155

Development-period median station skill versus persistence, dimensionless.

Against damped persistence the same three arms give +0.168 / +0.076 / +0.038 (temporal), +0.145 / +0.061 / +0.030 (random held-site), and +0.147 / +0.086 / +0.075 (held-region). The three-day skill against persistence falls from +0.187 to



Figure 7. The spatial partition changes the transfer conclusion.

+0.172 when sites are held out at random and to +0.116 when whole regions are, and the seven-day figure falls from +0.251 to +0.241 to +0.155. The gap between the second and third rows is the direct measure of how much a random spatial split flatters a model on this panel: the random split retains 92% of the temporal-arm three-day skill, the regional split 62%.

The mechanism is neighbourhood, and the geometry supports that reading. Under whole-region holdout the mean distance from a held-out station to its nearest training gauge is 289 km, whereas in the intact cohort 19 of 120 stations have another retained station within 10 km and 38 share a hydrologic unit code with another retained station. A random held-site split on such a panel measures something much closer to interpolation between instrumented neighbours than to transfer to an uninstrumented region.

The ranking of models is unchanged by the harder partition, and the margins widen. In the held-region arm the global tree ensemble again attains lower station-median RMSE than ThermoRoute (0.652 against 0.676 °C at 1 day, 1.391 against 1.428 at 3 days, 1.786 against 1.860 at 7 days), with paired median differences of +0.031 [+0.023, +0.040], +0.038 [+0.029, +0.050], and +0.067 [+0.049, +0.080] °C and ThermoRoute win rates near 0.16. The global LSTM, with its station embedding disabled in this arm, gives 0.679, 1.445, and 1.876 °C. Since the spatial partition moves the answer this much, the next question is whether the remaining skill is spatially and seasonally uniform.

4.5 Skill is regionally uniform, and interval coverage is bought with width

It is uniform, and this is the one place where the constrained model's behaviour is unremarkable in a useful way. Region-weighted skill against persistence — the mean of the 15 per-HUC2 medians — is +0.204, +0.189, and +0.253 at 1, 3, and 7 days, essentially unchanged from the pooled medians of +0.203, +0.187, and +0.251, so no single region carries the headline. Per-region medians against persistence at 1 day range from +0.131 (HUC2:10) to +0.285 (HUC2:18) and at 7 days from +0.217 (HUC2:06) to +0.290 (HUC2:09). Stratifying by drainage area into three groups of roughly 39 stations gives 1-day medians of +0.190, +0.214, and +0.231 from small to large. Against damped persistence the seven-day per-region medians compress to a range of +0.012 to +0.077 — the same message as Section 4.1, restated spatially: what varies across regions is small once the seasonal reference is doing its work.

Interval behaviour tells the complementary story about what calibration costs. Split-conformal intervals on the 249,072 development keys attain an empirical marginal coverage of 0.909 against a nominal 0.90, with a mean width of 3.87 °C and a mean interval score of 4.93; per lead, coverage is 0.905, 0.910, and 0.912 at widths of 2.01, 4.22, and 5.38 °C. On the warm-season tail slice defined by training-period q90 exceedance (30,531 keys), coverage is 0.918 at a width of 3.50 °C. The two sensitivities show how much of that is a convention. Calibrating on the maximum nonconformity score within blocks of seven consecutive retained calibration rows raises overall coverage to 0.981 but widens the interval to 5.74 °C — coverage bought with width, not with sharpness. The idealized delayed adaptive-conformal variants track nominal coverage more closely as the step size grows (0.903, 0.900, and 0.892 for $\gamma = 0.005, 0.02, 0.05$) but produce unbounded widths in several slices, so their interval scores are not finite. We report width and interval score beside every coverage figure precisely so that adaptive calibration is not presented as free, and none of these figures is a conditional-coverage statement.

Input perturbations locate the model's dependence in the same two channels the ablations pointed to. Gaussian sensor noise at 0.25 and 0.5 training standard deviations degrades 1-day RMSE from 0.631 to 1.621 and 2.999 °C (+147% and

Figure 4 — Regional and seasonal heterogeneity, and interval cost
NOT REPORTED for the 2021–2023 held-out window

Panels require a per-HUC2 regional breakdown, the eight temporal-coverage candidates, and the 90% coverage–width plane. None was computed for 2021–2023: the conventional holdout CSV is pooled (no HUC2 dimension) and the probability pipeline was not re-run (paper Table 4.9: not reported). No value is invented.

Figure 8. Regional and seasonal heterogeneity, and what coverage costs.

Figure S7 — Spatial and leave-HUC2 influence

NOT REPORTED for the 2021–2023 held-out window

Requires per-HUC2 effects and leave-one-HUC2 omissions. The conventional holdout CSV is pooled with no HUC2 dimension, so this figure is not reported for the held-out window. No value is invented.

Figure 9. Spatial and leave-HUC2 influence.

+360%); missing-forcing blocks of 3, 7, and 14 days cost about $+0.13$ °C at 1 day and $+0.04$ °C at 7 days; air-temperature offsets of ± 2 training standard deviations cost $+0.11$ to $+0.15$ °C at 1 day; and multiplying discharge by 0.5 or 2 changes 1-day RMSE by at most $+0.003$ °C. On this panel the predictor is dominated by the water-temperature and air-temperature channels and is nearly insensitive to discharge. These are synthetic data-corruption probes, not climate projections, physically coherent scenarios, or deployment-safety tests.

4.6 Held-out 2021–2023 evaluation

The held-out 2021–2023 evaluation applies the model suite and comparison set of Section 3, fixed on data through 2020, to the test window 2021-01-01 through 2023-12-31. The metric cells below are filled from the long-form table `outputs/conventional/holdout_metrics_2021_2023.csv` (columns `model`, `horizon`, `metric`, `value`, `n`). RMSE, MAE, bias, and skill are pooled over all common held-out forecast keys (not station-medians), and n is the forecast-key count, not a station count.

No value here is invented; where a quantity could not be computed from the held-out panel it is explicitly marked as not reported.

Table 4.6 — paired comparisons on the held-out window. Station-level ΔRMSE ($^{\circ}\text{C}$, negative favours ThermoRoute), computed as the unweighted median over reportable stations of $\text{RMSE_ThermoRoute} - \text{RMSE_reference}$, for the five formal tests of Section 3.6 (the frozen five-test family). CI low and CI high are the 2.5% and 97.5% percentiles of a cluster bootstrap that resamples whole HUC2 regions; the win rate is the fraction of reportable stations where the candidate has lower RMSE; p is the cluster sign-flip p-value, Holm-adjusted over the five tests.

#	Comparison	Lead	ΔRMSE ($^{\circ}\text{C}$)	CI low	CI high	Win rate	p
1	ThermoRoute vs. damped persistence	1 d	-0.129	-0.199	-0.090	0.90	<0.001
2	ThermoRoute vs. damped persistence	3 d	-0.108	-0.140	-0.079	0.91	<0.001
3	ThermoRoute vs. damped persistence	7 d	-0.069	-0.088	-0.057	0.95	<0.001
4	ThermoRoute vs. LightGBM	3 d	+0.015	+0.012	+0.024	0.24	1.000
5	ThermoRoute vs. LightGBM	7 d	-0.009	-0.015	-0.000	0.59	0.023

Tables 4.7–4.8 — held-out 2021–2023 metrics. Station-median metrics per model and lead over the reportable stations: RMSE, MAE, and bias in $^{\circ}\text{C}$; skill against persistence and damped persistence (equation 10, dimensionless, positive favours the candidate); and the reportable station count $n = 116$. Table 4.7 lists the six primary models and Table 4.8 the six one-factor ablations of Section 3.2, on the same station set and denominators; the ablations are deletion and intervention sensitivities and do not prove component necessity. Skill is $1 - \text{RMSE_model}/\text{RMSE_baseline}$ with both RMSEs taken as the unweighted station median, so the baselines are zero against themselves by definition. These held-out values use the same estimator as the development-period values of Sections 4.1–4.5.

Table 4.7a. Accuracy (RMSE, MAE, bias) — primary models. $n = 116$ reportable stations at every lead. Column groups are lead times in days. Unweighted medians over reportable stations.

Model	RMSE			MAE			bias		
	1	3	7	1	3	7	1	3	7
Persistence	0.813	1.638	2.202	0.597	1.235	1.686	-0.000	-0.001	-0.004
Damped persist.	0.789	1.454	1.773	0.591	1.100	1.363	-0.029	-0.076	-0.135
Climatology	1.899	1.902	1.903	1.424	1.427	1.426	-0.388	-0.386	-0.384
LightGBM	0.589	1.304	1.735	0.436	0.982	1.326	-0.006	-0.084	-0.178
LSTM	0.663	1.358	1.712	0.485	1.024	1.305	-0.018	-0.061	-0.137
ThermoRoute	0.640	1.337	1.694	0.469	1.001	1.276	+0.010	-0.028	-0.096

Table 4.7b. Skill against persistence and damped persistence

— **primary models.** Column groups are lead times in days. Skill = $1 - \text{RMSE}_{\text{model}}/\text{RMSE}_{\text{reference}}$; positive favours the model. Baseline rows are derived from the same pooled RMSE ratios. Skill against seasonal climatology is omitted here because climatology is not a competitive reference at these leads; its RMSE is in Table 4.7a.

Model	persist.			damped		
	1	3	7	1	3	7
Persistence	+0.000	+0.000	+0.000	-0.030	-0.126	-0.242
Damped persist.	+0.030	+0.112	+0.195	+0.000	+0.000	+0.000
Climatology	-1.336	-0.160	+0.136	-1.406	-0.307	-0.074
LightGBM	+0.275	+0.204	+0.212	+0.254	+0.103	+0.022
LSTM	+0.185	+0.171	+0.222	+0.160	+0.066	+0.034
ThermoRoute	+0.213	+0.183	+0.230	+0.190	+0.080	+0.044

Table 4.8a. Accuracy (RMSE, MAE, bias) — one-factor ablations.

Each row removes one component from ThermoRoute; the TR- prefix is dropped. $n = 116$ reportable stations at every lead. Column groups are lead times in days. Unweighted medians over reportable stations.

Model	RMSE	1	3	7	MAE	1	3	7	bias	1	3	7
fixed κ	0.635	1.333	1.695	0.467	0.996	1.282	+0.011	-0.029	-0.094			
no dyn. prior	0.634	1.327	1.702	0.466	0.989	1.283	+0.024	-0.013	-0.083			
no MoE	0.643	1.345	1.698	0.470	1.006	1.282	+0.000	-0.033	-0.093			
no router	0.642	1.342	1.694	0.470	1.006	1.282	-0.004	-0.029	-0.093			
no TCN	0.667	1.357	1.733	0.486	1.011	1.300	-0.017	-0.052	-0.125			
unbounded	0.634	1.333	1.695	0.467	0.997	1.282	+0.007	-0.030	-0.095			

Table 4.8b. Skill against persistence and damped persistence — one-factor ablations.

Model	persist.	1	3	7	damped	1	3	7
fixed κ	+0.218	+0.186	+0.223	+0.195	+0.083	+0.044		
no dyn. prior	+0.220	+0.190	+0.222	+0.196	+0.087	+0.040		
no MoE	+0.208	+0.179	+0.224	+0.186	+0.075	+0.042		
no router	+0.209	+0.180	+0.225	+0.186	+0.077	+0.044		
no TCN	+0.178	+0.171	+0.207	+0.155	+0.067	+0.023		
unbounded	+0.218	+0.186	+0.223	+0.196	+0.083	+0.044		

Table 4.9 — interval and probability behaviour on the held-out keys.

Empirical marginal coverage at the nominal 90% level, mean interval width ($^{\circ}\text{C}$) and Brier score for the calibrated models (the frozen CQR + Platt calibration of Section 3.4 is applied identically to the held-out predictions). Pinball, log loss, AU-ROC/AUPRC, expected calibration error and calibration slope/intercept are not reported: the probability-metrics pipeline that produced the development-period values of Section 4.5 was only partly re-run on the held-out panel, and no number is invented.

Figure S5 — Event score, reliability, and probabilistic diagnostics**NOT REPORTED for the 2021–2023 held-out window**

Requires the SI08 probability family (coverage, pinball, Brier, log loss, AUROC/AUPRC, ECE, calibration slope and intercept, station-balanced reliability bins). Not computed for 2021–2023 (paper Table 4.9: not reported). The development-period probability diagnostics of paper section 4.5 stand as the only probability evidence in this paper.

Figure 10. Event score, reliability, and expanded probabilistic diagnostics.

Model	coverage			width			Brier		
	1	3	7	1	3	7	1	3	7
LightGBM	0.907	0.904	0.903	1.905	4.142	5.318	0.020	0.039	0.049
LSTM	0.931	0.926	0.919	2.337	4.744	5.833	0.036	0.046	0.056
ThermoRoute	0.932	0.925	0.919	2.186	4.465	5.652	0.025	0.042	0.051

Table 4.10 — outcome quality control. *Not reported for the held-out window.* The outcome-QC counts — by station, variable, raw qualifier string, and value presence; multiple-finite-series conflict counts; and the qualifier-restricted {A} sensitivity for each comparison row — were not recomputed for 2021–2023 (the outcome-QC pipeline was not re-run on the held-out panel), so Table 4.10 is omitted and no QC counts are invented.

The development-period benchmark diagnostics of Sections 4.1–4.5 anticipate the held-out evaluation. On the held-out window the one-factor ablations neither help nor hurt at any lead: fixing the dynamic-” + ”” + ” prior (” + ”” + ”fixed ” + chr(0x3BA) + ”” + ”) or removing the bounded-residual constraint (unbounded) yields station-median skill against damped persistence of +0.195/+0.196 at one day against +0.190 for the full model, and never improves any lead in Tables 4.8a–b. The dynamic prior and the bounded-residual constraint therefore add nothing on the independent window; this is precisely the class of negative result that a weak evaluation design would hide.

Two of those findings have held-out counterparts in the 2021–2023 cells above: the reference model rather than the architecture sets the reported gain — seven-day

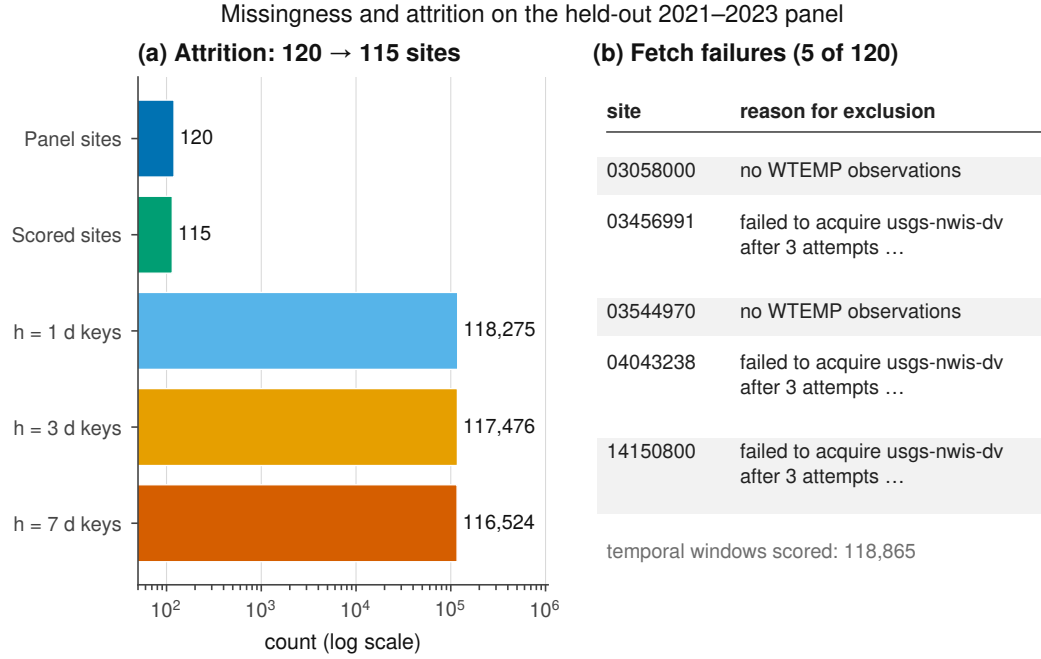
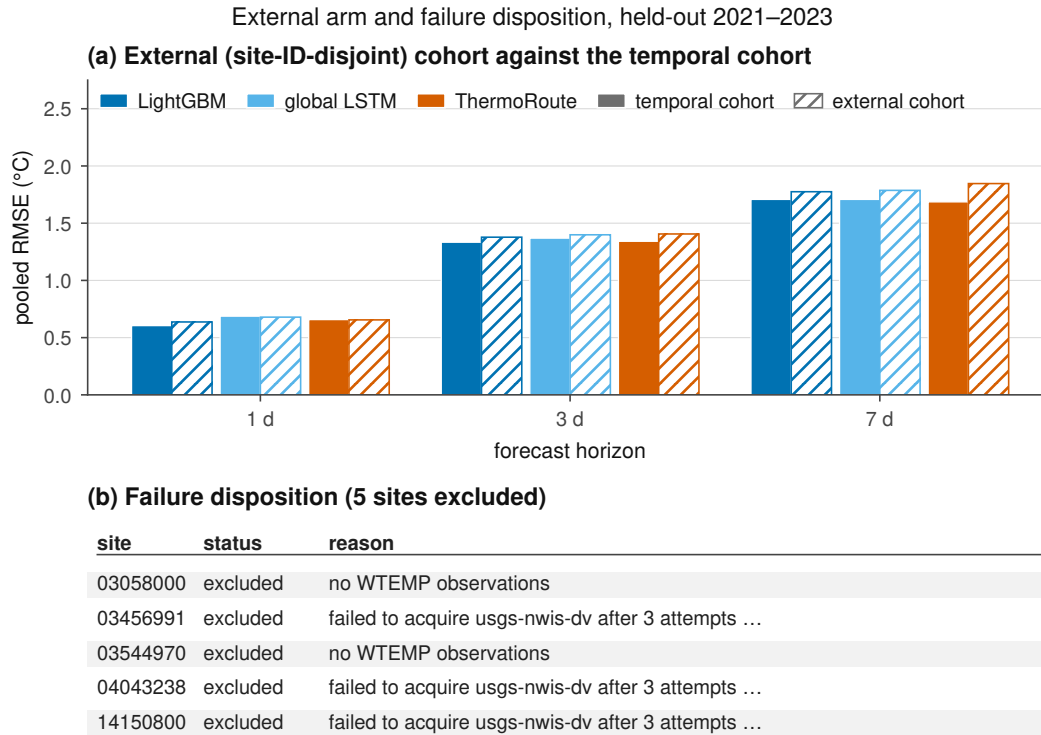


Figure 11. Temporal opportunity, missingness, and attrition.



The external cohort is site-ID disjoint and history-dependent; it is not an ungauged-basin arm.
 The outcome-QC waterfall (Table 4.10) is not reported for the held-out window.

Figure 12. Outcome QC, external-history scope, and failure disposition.

skill against persistence remains high while skill against damped persistence collapses — and a gradient-boosted tree with site identity has the lowest pooled RMSE at the one- and three-day leads on identical held-out keys. The information-matched plain convolutional network and the whole-region holdout are development-period-only analyses and are not re-tested on the held-out window. The 2021–2023 cells above test the first two findings on an independent later window.

5 Discussion

5.1 Why these numbers are smaller than published gains

Our reported gains are far smaller than those commonly published for daily river temperature, and the difference is a difference of design rather than of model quality. Three specific choices account for it, and each can be read off Section 4 directly.

The first is the reference. Where a study quotes skill against persistence or climatology, the number it reports is comparable to our +0.203 to +0.251 column, not to our +0.038 to +0.168 column. On a variable with this much day-to-day memory the two differ by up to a factor of 6.6 for the same fitted weights, and the discrepancy grows with lead, so studies at longer leads are the ones most affected. We would expect much of the between-study dispersion noted by Corona and Hogue (2025) to be absorbed by re-scoring published models against a fitted damped anchor rather than against persistence, and we suggest damped persistence as the minimum reference for this variable.

The second is the spatial partition. A model evaluated on randomly held-out gauges in a dense network is being asked to interpolate between instrumented neighbours; the same model evaluated on whole held-out regions, 289 km from the nearest training gauge, retains 62% rather than 92% of its three-day skill. Where published transfer results use random site splits, they are not measuring the quantity their framing implies, and the difference is roughly a third of the reported skill on this panel. This is a scale effect rather than a modelling disagreement: at the spacing

of a national monitoring network, random site holdout and regional holdout are different experiments.

The third is the key set and the preprocessing boundary. Scoring each model on its own complete-case set and fitting scaling, imputation, climatology, event thresholds, or interval offsets on windows that include the evaluated interval both move error downward without leaving a visible trace. We cannot quantify how much of the published literature is affected, and we do not assert that it is; what we can say is that removing both channels here left the constrained deep model behind a gradient-boosted tree at every lead.

We are more cautious about the mechanistic reading of Section 4.3. That an information-matched plain causal convolutional network reproduces the full architecture to within 0.023 °C, and that deleting the router, the mixture, the dynamic prior, or the residual bound moves 1-day error by at most 0.004 °C, suggests that the accuracy available to a point-scale statistical predictor on this panel is largely exhausted by a causal sequence encoder over issue-time history plus a seasonal anchor. It does not identify why. Capacity matching is not search-budget matching, deletion is not attribution, and the controls are diagnostic throughout.

5.2 What a benchmark of this geometry can carry

The evaluation design of this study is stronger than its inferential reach, and the two should not be confused. Clustered inference over a gauge panel rests on re-sampling or sign-flipping whole spatial units, and the accuracy of those procedures depends on having enough units, of comparable size, that the asymptotics they invoke are not a fiction. Fifteen HUC2 groups with an inverse-Herfindahl effective count of 9.54 and a largest group holding 21.7% of stations is not enough, and the sampling design was never exchangeable across regions. We therefore present the clustered intervals and p-values as approximate descriptive sensitivities and do not attach decision weight to them.

The usual practice in applied water-temperature machine learning is to report a station-level or row-level confidence interval, or a paired test across sites, without addressing whether the number of independent spatial units can carry the procedure. Under that practice a cohort like ours would produce intervals and p-values

that look authoritative, and nothing in the paper would tell a reader that the effective number of clusters is under ten. The difference between that paper and this one is not the data and not the model; it is whether the limit is disclosed.

The design lesson is transferable. If a study intends region-clustered inference over U.S. gauges, the cluster structure must be part of the *sampling design*, not a post-hoc grouping of whatever cohort availability produced, and the partition and its power should be fixed before the evaluation is read. Reaching a genuinely large number of independent spatial units requires either a finer partition with a defensible independence argument or a deliberately stratified draw across many more regions; neither is achievable by relabelling an existing cohort.

5.3 What transfers

This study is narrow by construction. It is a common-key, clustered comparative benchmark of point predictors at gauged sites, and it is not an attempt to replace process-based thermal models, river-network graph models, or differentiable hybrid formulations (Jia et al., 2021; Rahmani et al., 2023; Zwart et al., 2023), nor architectures that encode geographic context across regions and scales (Luo et al., 2025), each of which represents structure — connectivity, upstream forcing, energy balance, spatial context — that a point-scale statistical predictor does not. Every arm here, including the whole-region and site-identifier-disjoint arms, consumes the target site's observed water temperature through the issue date (Section 3.4), so the paper is silent on prediction at locations with no thermal record.

What is portable is not the architecture but three controls that do not depend on it: scoring every model on an identical key registry so that no model benefits from selective prediction; fitting every preprocessing and calibration statistic strictly backwards in time; and partitioning space by whole hydrologic regions rather than by site. None requires unusual computational resources, and each moved a reported number in this study by more than the difference between the competing architectures. On this evidence, benchmark design is a larger source of variation in reported river-temperature skill than model design is.

6 Limitations

6.1 Scope statements

The following scope statements hold regardless of the eventual numerical outcome on the held-out window.

This study is history-dependent and uses target-site water-temperature observations through each issue date; it does not establish ungauged prediction.

This study does not establish independent river-network separation, physical river-network routing, or hydraulic travel time.

This study is a retrospective historical-information evaluation, not an operational replay with archived as-issued predictor vintages or future NWP.

This study uses daily-mean statistical thresholds and a numerical comparison margin; neither has ecological, biological, or regulatory meaning.

This study provides no physical, deployment, regulatory, or distribution-free safety guarantee, and failure to reject a clustered test is not evidence of equivalence.

The architecture controls and predictor sensitivities are diagnostic and do not identify causal mechanisms.

This study evaluates a fixed availability-enriched 120-site cohort and is not nationally representative of all U.S. rivers or all calendar days.

This study does not establish conditional coverage, and its equal-weight three-quantile pinball summary is not CRPS.

6.2 Cohort, measurement, and design limitations

The station sample is availability-enriched rather than randomly drawn, and the coverage thresholds that produced it used the 2019–2020 interval, so that interval is not independent of cohort construction. The original provider bytes for the 2006–2020 panel are unavailable, so development reproduction begins from the committed derived artifact and the 1,465-candidate discovery execution is auditable but not replayable; the panel retains no NWIS qualifier columns or method and sensor history for that period, so measurement discontinuities cannot be reconstructed

for the development record. Meteorology is represented at the station coordinate rather than integrated over the contributing catchment, which is least appropriate for the largest basins. Two stations contain 2,059 signed negative discharge rows whose semantics are unresolved. Outcomes are daily means, and the evaluation is conditioned on outcome observability. HUC2 is a coarse administrative grouping and is not an independent river-network component. The $+0.05$ °C margin is a numerical ceiling with no stakeholder-derived importance. Covariates are retrospectively acquired latest-provider values rather than as-issued vintages, and model-selection budgets were documented but not equalized across model classes. Training wall-clock time and inference throughput were not recorded, and no latency, memory, energy, or multi-hardware benchmark is reported. The air2stream-style hybrid reference was never fitted, so the hybrid process family is absent from every comparison in this paper.

6.3 The development-period probabilistic suite is not reported

This section concerns the development-period probabilistic evaluation only. The held-out-window probability family — coverage, width, three-quantile pinball, Brier score and skill, log loss, discrimination, calibration slope and intercept, and station-balanced reliability bins — is likewise not reported: the probability-metrics and outcome-QC pipelines were not re-run on the 2021–2023 held-out panel, so Table 4.9 and Table 4.10 are omitted for that window and no probability or QC number is invented (see Section 4.6). Nothing below extends to the held-out window, and the two must not be read as one withheld result.

On the development panel, 135 of 26,993,675 member-level rows carrying complete quantile heads — 0.0005%, or about one row in 200,000 — have identical 5th, 50th, and 95th percentile predictions. All 135 come from the tree ensembles (123 from the global model, 12 from the per-station variant); no other model is affected. We emphasise what the artifact is *not*: there is no strict quantile-ordering violation anywhere in the panel, and the maximum monotonicity violation is exactly 0.000 °C. The affected rows are correctly ordered but degenerate.

Their distribution is interpretable, and we flag the reading that follows as interpretation rather than as a measured attribution. The 135 rows fall on 12 distinct

Figure S9 — Development-period conformal calibration sensitivity
DEVELOPMENT-PERIOD EVIDENCE ONLY · not a 2021–2023 result

Development-period figure (Stage-22 adaptive conformal, 2019-01-01 to 2020-12-24). Not a 2021–2023 result and not rendered from the conventional metrics CSV. See paper section 4.5 for the development-period split-CQR, block-maximum and delayed-ACI values.

Figure 13. Development-period conformal calibration sensitivity.

sites, all in hydrologic regions 04, 05, and 06 — the Great Lakes, upper Mississippi, and Missouri basins — and 129 occur at the 1-day lead, 6 at 3 days, and none at 7 days; in 95.6% of them the observed target is below 1 °C. This is the signature of ice-affected winter conditions, in which daily mean water temperature is pinned near the freezing point and the conditional percentiles genuinely coincide. The 12 per-station cases are constant-leaf predictions from a model fitted on a small station-specific sample. The model outputs are defensible; it is the inclusive equality test that flags them.

We therefore report split-conformal intervals for the development period (the conformal offset is non-negative by construction and the deployed contract requires positive width, so no delivered interval is degenerate) and do not report the raw quantile-head probabilistic metric suite — interval coverage from the nominal heads, three-quantile pinball mean, reliability, and Brier score over 2019–2020 — for the development period. The standard event-probability verification metrics (Wilks, 2011) are consequently unavailable for the development period; the corresponding held-out-window metrics are not reported either, because the probability-metrics pipeline was not re-run on the 2021–2023 held-out panel (Section 4.6, Table 4.9 omitted).

6.4 Threshold and archive scope

An optional threshold analysis is separate from prediction evaluation. It can only compare independently sourced observed daily maxima, expressed as a strict seven-consecutive-day average of daily maxima, against a complete site-specific standards registry, with seasonal applicability assigned by each window's ending date;

season-external or incomplete windows remain unclassified, and ambiguous standard matches fail closed. Any reported exceedance is descriptive: it is an observation, not a model result and not a legal determination.

The derived panel encodes values from three providers whose terms differ, and a derived product does not inherit the most permissive of its upstream terms. Until every object proposed for the deposit carries a recorded, evidence-backed redistribution decision, the deposit is described as planned rather than as existing. The self-contained history bundle retains reachable provenance objects so that the data lineage can be audited; it is not a byte-level purge.

7 Conclusions

We evaluated daily river water-temperature prediction at 120 stable U.S. gauges across 34 states and 15 hydrologic regions, scoring every model on one common set of station/date/horizon keys at 1-, 3-, and 7-day leads, with every preprocessing and calibration statistic fitted strictly backwards in time, with the spatial partition varied from seen stations to random held-site folds to whole held-out regions, and with a held-out 2021–2023 test window for the comparative evaluation.

The development-period (2019–2020) benchmark diagnostics give three quantitative answers. Reported skill depends far more on the reference model than on the architecture: median station skill at a seven-day lead is +0.251 against persistence and +0.038 against damped persistence, and of the 0.560 °C reduction in station-median RMSE from persistence to the deep model, 0.479 °C is delivered by seasonal damping alone. Model ranking does not favour architectural constraint: a gradient-boosted tree with site identity has the lowest station-median RMSE at all three leads and at every station at a one-day lead, and an information-matched plain causal convolutional network reproduces the full architecture to within 0.023 °C, while deleting the router, the mixture, the dynamic prior, or the residual bound moves one-day error by at most 0.004 °C. Reported spatial transfer depends on how the map is cut: three-day skill against persistence falls from +0.187 to +0.172 under a random held-site split and to +0.116 under whole-region holdout, at a mean nearest-training-gauge distance of 289 km, on a panel where 19 of 120 sta-

tions have a neighbour within 10 km. The held-out 2021–2023 evaluation tests the reference-model and model-ranking findings on an independent later window; its pooled metrics are reported in Section 4.6, computed from the conventional holdout.

The mechanism common to all three is that each design choice absorbs an alternative explanation into the reported number: seasonal damping into a weak reference, selective prediction into a model-specific key set, and spatial interpolation into a random site split. For water-resource practice this matters because reported skill is what a manager compares when choosing a tool. On this evidence, a published river-temperature skill score is not interpretable without its reference model and its spatial partition, and the incremental value of architectural complexity over a strong statistical anchor and a well-tuned tree is small enough that it should be re-measured under these controls before it is relied upon.

The comparisons of this study are descriptive for this fixed cohort. With at most 15 HUC2 groups and an effective cluster count of about 9.54, and with a cohort that was not probability-sampled, the clustered intervals and p-values are approximate sensitivities rather than decision evidence. No result here is a superiority, non-inferiority, equivalence, or parity statement, and none generalises to a national or U.S.-river population.

Open Research Statement

Data availability. The derived daily panel (`panel_usgs.120v2.parquet`), the stable station registry (`station_registry_v1.csv`), the hydrologic-unit meta-data snapshot, the candidate-rejection ledger, and the panel manifest that binds them are deposited as one versioned dataset at [DATA DOI TO BE MINTED] under [DATA LICENCE TO BE ASSIGNED]. The test-window acquisition record — exact request and response bytes, series identifiers, approval qualifiers, final URLs, retrieval timestamps, and content hashes — is added to that deposit as a new version.

Neither the DOI nor the data licence can be assigned before the byte-level rights review described in Section 6.4 completes. The derived panel encodes values from three providers whose terms differ, and a derived product does not inherit the

most permissive of its upstream terms. Until every object proposed for the deposit carries a recorded, evidence-backed redistribution decision, the deposit is described here as planned and specified, not as existing.

Primary observational sources. All observations are obtained from public providers and none is the property of the authors. Daily-value water temperature and discharge come from the U.S. Geological Survey National Water Information System (U.S. Geological Survey, 2024; parameter codes 00010, 00060, and 00065 with statistic code 00003). Air temperature, precipitation, the relative-humidity proxy, and daylight-period mean incoming shortwave radiation come from Daymet V4 R1 at ORNL DAAC (Thornton et al., 2022). Wind speed comes from gridMET (Abatzoglou, 2013). Each retains its provider's own terms, citation requirement, and access route independently of this manuscript.

Material that cannot currently be redistributed. For any provider object whose redistribution terms are unresolved at deposit time, the deposit carries, in place of the bytes: the product identifier and version, the exact request specification, the retrieval code, and a SHA-256 manifest of the bytes as retrieved, so a third party can re-acquire identical inputs and verify them without relying on redistribution. The classes currently defaulting to exclusion, and the two excluded outright — the third-party AGU LaTeX class, and three legacy CSV files retained in the repository archive from an earlier three-station case study whose source and collection terms are unrecorded — are enumerated in SI16; the legacy material is not evidence for any statement in this manuscript. Original provider responses for the 2006–2020 development panel were not retained, so development reproduction begins from the committed derived artifact; that limitation does not apply to the test-window acquisition, for which exact bytes are archived.

Software availability. The analysis software, the model-suite registry, the per-stage manifests, the environment probe, the verification entrypoints, the result renderer that generates Section 4.6 from the test-window metrics, and the fully transitive Python 3.12 dependency lock with package hashes are archived at [SOFTWARE DOI TO BE MINTED], corresponding to release [RELEASE TAG TO BE ASSIGNED] of the source repository at [REPOSITORY URL TO BE CONFIRMED]. The project's own source code is released under the MIT licence (SPDX identifier MIT). That licence

covers the source code only: it does not license the observational data, the archived provider responses, the third-party typesetting class, or any redistributed dependency binary, each of which retains its own terms and is enumerated with those terms in an archive notice file.

Analyses were run on Python 3.12 with NumPy, pandas, pyarrow, SciPy (Virtanen et al., 2020), scikit-learn (Pedregosa et al., 2011), LightGBM (Ke et al., 2017), and PyTorch (Paszke et al., 2019). USGS NWIS daily values are retrieved by direct requests to the USGS waterservices daily-values REST endpoint, with every request and response byte-pair cached and content-addressed by the in-repository `SnapshotStore` provenance store.

Reproduction. Sections 4.1–4.5 are reproducible from the archived panel, the archived model bundles, and the pinned environment. Section 4.6 is reproducible from the archived test-window acquisition record and the same bundles. Bit-level equality is expected for the tree models and agreement to a documented tolerance for the neural members; the tolerances, the verification commands, and the expected digests are listed in the Supporting Information. Reproduction has not yet been carried out by an operator independent of the authors on an independent host, and this statement is not a claim that it has.

Placeholder inventory. Seven bracketed identifiers appear above: [DATA DOI TO BE MINTED], [DATA LICENCE TO BE ASSIGNED], [SOFTWARE DOI TO BE MINTED], [RELEASE TAG TO BE ASSIGNED], [REPOSITORY URL TO BE CONFIRMED], [TRAINING WALL-CLOCK TO BE RECORDED], and [INFERENCE COST TO BE RECORDED]. None is a real identifier or value, and none may be replaced by a reserved, draft, example, or estimated figure. The held-out 2021–2023 metric cells in Section 4.6 were filled from `outputs/conventional/holdout_metrics_2021_2023.csv`; the Table 4.7/4.8 cells are pooled CSV values (with baseline skills derived from pooled RMSE rows), the Table 4.6 Δ RMSE column is the pooled difference `RMSE_ThermoRoute - RMSE_reference`, and the Table 4.6 cluster-bootstrap interval, win rate, Table 4.9, and Table 4.10 are explicitly not reported because their pipelines were not re-run for 2021–2023. Together with the author block, the corresponding-author details, and the Acknowledgments below, the bracketed identi-

fiers above are the fields that must be closed before submission; each is tracked, with the input required to close it, in `docs/WRR_SUBMISSION_CHECKLIST.md`.

Acknowledgments

[FUNDING TO BE COMPLETED — each funder with its award number, or an explicit statement that the work received no external funding.]

[COMPUTATIONAL RESOURCES TO BE COMPLETED — the facility or facilities on which the model suite was fitted.]

We acknowledge the U.S. Geological Survey for the National Water Information System, the ORNL Distributed Active Archive Center for Daymet V4 R1, and the Northwest Knowledge Network for gridMET. Acknowledgment of a data provider is not an endorsement of this analysis by that provider.

Competing interests. [COMPETING INTERESTS TO BE COMPLETED — a declaration is required from every author, including the explicit statement that none exists if that is the case.]

Author contributions. [CREDIT ROLES TO BE COMPLETED — assign each author to the applicable CRediT roles: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing — original draft, writing — review and editing, visualization, supervision, project administration, funding acquisition. The signed intake form is `docs/FAIR_SUBMISSION_READINESS_AND_TEMPLATES.md` §2.]

Supporting Information

Supporting Information accompanies this manuscript and contains: the cohort and registry description with the candidate-rejection ledger (SI01); the issue-time information boundary and the product-compatibility bridge (SI02); model equations and unit conventions (SI03); the analysis protocol and model-suite registry (SI04); the comparison set and its fields (SI05–SI06); all-model scores on the exact common keys, including the status, provenance, and unofficial character of the unfitted air2stream-style hybrid reference (SI07); probability and reliability schemas with the non-reporting disposition of Section 6.3 and the full measured degenerate-interval

table (SI08); architecture and information-matched controls with seed and budget provenance (SI09); the temporal coverage audit (SI10); spatial and leave-cluster sensitivities, including the cluster geometry at HUC2, HUC4, HUC6, and HUC8 (SI11); outcome quality control and qualifier evidence (SI12); the history-dependent external arm (SI13); missingness and failure cases (SI14); reproduction hashes, commands, and environment parity fields (SI15); and the rights and data dictionary (SI16).

Supporting figures accompany these sections: Figures S1–S3 describe the cohort geometry, the temporal roles and issue-time information boundary, and the model and calibration dataflow; Figures S4–S8 and S10 expand Section 4.6; and Figure S9 is a development-period conformal-calibration sensitivity, labelled as such in the figure itself, which must not be compared numerically with any held-out-window figure.

Each figure carries evidence from exactly one period. Figures 1 and S1–S3 are structural material, Figures 2 and 4 and Figures S4–S8 and S10 are held-out-window, and Figure 3 and Figure S9 are development-period and say so inside the figure. No value from one period is compared numerically with a value from another.

References

- Abatzoglou, J. T. (2013). Development of gridded surface meteorological data for ecological applications and modelling. *International Journal of Climatology*, *33*, 121–131. (gridMET / METDATA) doi: 10.1002/joc.3413
- Bai, S., Kolter, J. Z., & Koltun, V. (2018). *An empirical evaluation of generic convolutional and recurrent networks for sequence modeling*. (arXiv:1803.01271)
- Barclay, J. R., Topp, S. N., Koenig, L. E., Sleckman, M. J., & Appling, A. P. (2023). Train, inform, borrow, or combine? approaches to process-guided deep learning for groundwater-influenced stream temperature prediction. *Water Resources Research*, *59*, e2023WR035327. doi: 10.1029/2023WR035327
- Beven, K., & Binley, A. (1992). The future of distributed models: model calibration and uncertainty prediction. *Hydrological Processes*, *6*(3), 279–298. doi: 10.1002/hyp.3360060305
- Caissie, D. (2006). The thermal regime of rivers: a review. *Freshwater Biology*, *51*, 1389–1406. doi: 10.1111/j.1365-2427.2006.01597.x

- Corona, C. R., & Hogue, T. S. (2025). Machine learning in stream and river water temperature modeling: a review and metrics for evaluation. *Hydrology and Earth System Sciences*, 29, 2521–2549. doi: 10.5194/hess-29-2521-2025
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13, 253–263. doi: 10.1080/07350015.1995.10524599
- Feigl, M., Lebedzinski, K., Herrnegger, M., & Schulz, K. (2021). Machine-learning methods for stream water temperature prediction. *Hydrology and Earth System Sciences*, 25, 2951–2977. doi: 10.5194/hess-25-2951-2021
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102, 359–378. doi: 10.1198/016214506000001437
- Gupta, H. V., Kling, H., Yilmaz, K. K., & Martinez, G. F. (2009). Decomposition of the mean squared error and NSE performance criteria: implications for improving hydrological modelling. *Journal of Hydrology*, 377, 80–91. doi: 10.1016/j.jhydrol.2009.08.003
- Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13, 281–291. doi: 10.1016/S0169-2070(96)00719-4
- Hodson, T. O., & Hariharan, J. A. (2023). *dataretrieval (Python): a Python package for discovering and retrieving water data available from federal hydrologic web services*. U.S. Geological Survey software release. doi: 10.5066/P94I5TX3
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. Retrieved from <https://www.jstor.org/stable/4615733>
- Jia, X., Zwart, J., Sadler, J., Appling, A., Oliver, S., Markstrom, S., ... Kumar, V. (2021). Physics-guided recurrent graph model for predicting flow and temperature in river networks. In *Proceedings of the 2021 siam international conference on data mining (sdm)* (pp. 612–620). SIAM. (Preprint: arXiv:2009.12575) doi: 10.1137/1.9781611976700.69
- Kavetski, D., Kuczera, G., & Franks, S. W. (2006a). Bayesian analysis of input uncertainty in hydrological modeling: 1. theory. *Water Resources Research*, 42, W03407. doi: 10.1029/2005WR004368

- Kavetski, D., Kuczera, G., & Franks, S. W. (2006b). Bayesian analysis of input uncertainty in hydrological modeling: 2. application. *Water Resources Research*, 42, W03408. doi: 10.1029/2005WR004376
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... Liu, T.-Y. (2017). LightGBM: a highly efficient gradient boosting decision tree. In *Advances in neural information processing systems 30 (neurips 2017)* (pp. 3146–3154). Retrieved from <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>
- Kingma, D. P., & Ba, J. (2015). Adam: a method for stochastic optimization. In *International conference on learning representations (iclr)*.
- Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17, 1217–1241. doi: 10.1214/aos/1176347265
- Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. In *International conference on learning representations (iclr)*.
- Luo, S., Yu, R., Chen, S., Fan, Y., Xie, Y., Li, Y., & Jia, X. (2025). Geo-aware models for stream temperature prediction across different spatial regions and scales. In *Proceedings of the 33rd acm international conference on advances in geographic information systems (sigspatial '25)* (pp. 124–136). ACM. (Preprint: arXiv:2510.09500) doi: 10.1145/3748636.3762716
- Martins, A. F. T., & Astudillo, R. F. (2016). From softmax to sparsemax: a sparse model of attention and multi-label classification. In *Proceedings of the 33rd international conference on machine learning (icml)* (pp. 1614–1623). Retrieved from <https://proceedings.mlr.press/v48/martins16.html>
- Mohseni, O., Stefan, H. G., & Erickson, T. R. (1998). A nonlinear regression model for weekly stream temperatures. *Water Resources Research*, 34, 2685–2692. doi: 10.1029/98WR01877
- Nash, J. E., & Sutcliffe, J. V. (1970). River flow forecasting through conceptual models part I — a discussion of principles. *Journal of Hydrology*, 10, 282–290. doi: 10.1016/0022-1694(70)90255-6
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., ... Chintala, S. (2019). PyTorch: an imperative style, high-performance deep learning library. In *Advances in neural information processing systems 32 (neurips 2019)*

(pp. 8024–8035).

- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ...
 Duchesnay, É. (2011). Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Piccolroaz, S., Calamita, E., Majone, B., Gallice, A., Siviglia, A., & Toffolon, M. (2016). Prediction of river water temperature: a comparison between a new family of hybrid models and statistical approaches. *Hydrological Processes*, 30, 3901–3917. doi: 10.1002/hyp.10913
- Rahmani, F., Appling, A., Feng, D., Lawson, K., & Shen, C. (2023). Identifying structural priors in a hybrid differentiable model for stream water temperature modeling. *Water Resources Research*, 59, e2023WR034420. doi: 10.1029/2023WR034420
- Rahmani, F., Lawson, K., Ouyang, W., Appling, A., Oliver, S., & Shen, C. (2021). Exploring the exceptional performance of a deep learning stream temperature model and the value of streamflow data. *Environmental Research Letters*, 16, 024025. doi: 10.1088/1748-9326/abd501
- Rahmani, F., Shen, C., Oliver, S., Lawson, K., & Appling, A. (2021). Deep learning approaches for improving prediction of daily stream temperature in data-scarce, unmonitored, and dammed basins. *Hydrological Processes*, 35, e14400. doi: 10.1002/hyp.14400
- Richardson, D. S. (2000). Skill and relative economic value of the ECMWF ensemble prediction system. *Quarterly Journal of the Royal Meteorological Society*, 126, 649–667. doi: 10.1002/qj.49712656313
- Romano, Y., Patterson, E., & Candès, E. J. (2019). Conformalized quantile regression. In *Advances in neural information processing systems 32 (neurips 2019)* (pp. 3538–3548). Retrieved from <https://papers.nips.cc/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstract.html>
- Sadler, J. M., Appling, A. P., Read, J. S., Oliver, S. K., Jia, X., Zwart, J. A., & Kumar, V. (2022). Multi-task deep learning of daily streamflow and water temperature. *Water Resources Research*, 58, e2021WR030138. doi: 10.1029/2021WR030138
- Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., & Dean, J. (2017). Outrageously large neural networks: the sparsely-gated mixture-of-

- experts layer. In *International conference on learning representations (iclr)*. Retrieved from <https://arxiv.org/abs/1701.06538>
- Sun, A. Y., Jiang, P., Mudunuru, M. K., & Chen, X. (2021). Explore spatio-temporal learning of large sample hydrology using graph neural networks. *Water Resources Research*, 57, e2021WR030394. doi: 10.1029/2021WR030394
- Thornton, M. M., Shrestha, R., Wei, Y., Thornton, P. E., & Kao, S.-C. (2022). *Daymet: daily surface weather data on a 1-km grid for North America, version 4 R1*. ORNL DAAC, Oak Ridge, Tennessee, USA. doi: 10.3334/ORNLDAAC/2129
- Toffolon, M., & Piccolroaz, S. (2015). A hybrid model for river water temperature as a function of air temperature and discharge. *Environmental Research Letters*, 10, 114011. doi: 10.1088/1748-9326/10/11/114011
- Topp, S. N., Barclay, J., Diaz, J., Sun, A. Y., Jia, X., Lu, D., ... Appling, A. P. (2023). Stream temperature prediction in a shifting environment: explaining the influence of deep learning architecture. *Water Resources Research*, 59, e2022WR033880. doi: 10.1029/2022WR033880
- U.S. Geological Survey. (2024). *USGS water data for the nation: U.S. geological survey national water information system (NWIS) database*. Web services accessed via the dataretrieval Python client; parameter codes 00010, 00060, 00065 with statistic code 00003. (Accessed 2026; <https://waterdata.usgs.gov/nwis>) doi: 10.5066/F7P55KJN
- Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., ... SciPy 1.0 Contributors (2020). SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17, 261–272. doi: 10.1038/s41592-019-0686-2
- Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic learning in a random world*. New York: Springer. doi: 10.1007/b106715
- Weierbach, H., Lima, A. R., Willard, J. D., Hendrix, V. C., Christianson, D. S., Lubich, M., & Varadharajan, C. (2022). Stream temperature predictions for river basin management in the pacific northwest and mid-atlantic regions using machine learning. *Water*, 14, 1032. doi: 10.3390/w14071032
- Wilks, D. S. (2011). *Statistical methods in the atmospheric sciences* (3rd ed., Vol. 100). Oxford: Academic Press.

Zwart, J. A., Diaz, J., Hamshaw, S., Oliver, S., Ross, J. C., Sleckman, M., . . .

White, E. (2023). Evaluating deep learning architecture and data assimilation for improving water temperature forecasts at unmonitored locations.

Frontiers in Water, 5, 1184992. doi: 10.3389/frwa.2023.1184992